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Laser powder bed fusion of zinc-based biomaterials: Process-microstructure-property relationships for biomedical applications

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Abstract

Additive manufacturing (AM), particularly laser powder bed fusion (L-PBF), offers unprecedented capabilities for fabricating complex, patient-specific three-dimensional (3D) structures with high geometric precision and design flexibility. These advantages are particularly valuable in applications that demand intricate architecture and functional integration, such as aerospace, automotive, biodegradable batteries, multifunctional composites, and, most notably, biomedical implants. Zinc (Zn) and its alloys have emerged as promising biodegradable metallic materials for biomedical applications, owing to their favorable cytocompatibility and moderate degradation rates, positioned between those of magnesium (Mg) and iron (Fe). Zn-based materials are particularly attractive for temporary load-bearing implants and cardiovascular devices, where controlled bioresorption is critical. Recent developments in selective laser melting, a form of L-PBF for processing Zn and its alloys, demonstrate substantial progress in tailoring microstructural and functional properties for medical applications. However, the successful fabrication of Zn-based components via L-PBF remains challenging due to Zn's low boiling point, high vapor pressure, and sensitivity to oxidation at elevated temperatures. These factors lead to process-induced defects, compositional instabilities, and variable degradation behavior, necessitating a deeper understanding of the interrelationships among process, structure, properties, and degradation. This review critically evaluates the current state of research on pure Zn and Zn-based alloys fabricated via L-PBF, focusing on process parameter optimization, metallurgical defects, microstructural control, mechanical performance, and in vitro and in vivo degradation behavior. Findings across these domains are integrated to highlight the fundamental challenges, consolidate the fragmented knowledge base, and outline future research directions to accelerate the clinical translation of Zn-based biomaterials. This effort is crucial for realizing the full potential of AM-enabled, bioresorbable metallic implants in next-generation regenerative medicine.

Keywords: additive manufacturing; Laser powder bed fusion; Zinc; Biodegradable and Biocompatible metal.

1. Introduction

Additive manufacturing (AM) is defined as the incremental construction of layers of material to build a 3D object from a computer-aided design (CAD) file. Rapid manufacturing (RM) and rapid prototyping (RP) are widely used terms for AM [1]. As stated by the American Society for Testing and Materials (ASTM), AM is a technique for creating objects from 3D CAD data, typically by stacking or building them layer by layer, in contrast to traditional subtractive manufacturing methods [2]. AM facilitates the assembly of multiple parts into a single component, providing advantages in weight reduction, fewer processing steps, lower manufacturing costs, and reduced complexity [3]. AM has gained attention due to its capacity for complex geometric design and greater flexibility, enabling shorter lead times [4]. It is capable of building complex parts with greater degrees of freedom [5]. Over the years, AM has outperformed conventional manufacturing in terms of accuracy, cost, speed, reliability, and the ability to produce complex products [6]. The buy-to-fly ratio is significantly lower than in conventional processes, indicating a 1:1 intake-to-end-product weight ratio in AM [7]. AM processes have reduced total costs by 0.3-0.5 times compared to traditional manufacturing processes [8]. According to ASTM, AM is broadly classified into seven major manufacturing processes, as shown in Fig. 1(a) [9]. Powder bed fusion (PBF) is one of the main AM techniques, primarily including the L-PBF subtype. It is a promising technique for producing complex, porous structures with a high surface finish. The basic raw materials for the process are metal powder or its mixture, which is spread over the substrate or base plate, followed by directing a high-energy laser beam over the powder to melt and fuse it to the preceding layers. This results in a high-density compact material with a smooth surface finish, offering better quality than traditional manufacturing methods [10]. PBF is the most widely used AM process for producing metal parts. It utilizes high-energy-density beams, such as lasers (L-PBF) and electron beams (EB-PBF), to fuse and melt the metal in the specified scan areas according to the CAD file, as illustrated in Figs. 1(b) and 1(c) [11]. Once the melting of the selected scan regions in a layer is complete, the build platform or bed is lowered to a predefined distance, and the subsequent layer of powder is placed over the entire bed. This process continues until the desired object or workpiece is produced [12]. L-PBF is receiving significant attention due to its ability to create intricate geometries and reduced surface roughness [13]. It provides high-dimensional accuracy and greater flexibility in materials and shapes while building complex components [14-16]. L-PBF can fabricate porous structures by creating micro and macro structures with exceptional performance and control over shape [17,18]. It offers advantages over traditional methods, such as powder metallurgy and casting, by allowing alterations to the material's properties through advancements in grain structure, reduced element segregation, and extended solution limits, thereby improving mechanical strength and compatibility [19-21]. Numerous materials, such as titanium alloy [22,23], nickel-based superalloy [24], aluminum alloy [25], magnesium alloy [26,27], zinc alloy [28], metallic glass [29], and shape memory alloy [30], have been produced through the widespread use of L-PBF. However, several obstacles hinder the widespread industrial and medical use of L-PBF. One challenge is the lack of technological standards or certification methods ensuring printing consistency,

influenced by various elements, products, powders, process variables, and post-treatment on the forming quality and properties [31]. The characteristics of L-PBF are highly reliant on grain orientation determined by the building direction [32,33]. In L-PBF, repeated thermal cycles during processing induce directional grain growth, resulting in an anisotropic microstructure [34]. The unique properties of L-PBF contribute to an anisotropic microstructure with non-uniform deformation under loading [35]. The scope of the review paper is outlined in Fig. 2, comprising basic outlines related to Zn-based biomedical implants, including biomedical metallic materials, traditional methods, microstructures and properties, essential requirements, biocompatibility, and future trends.

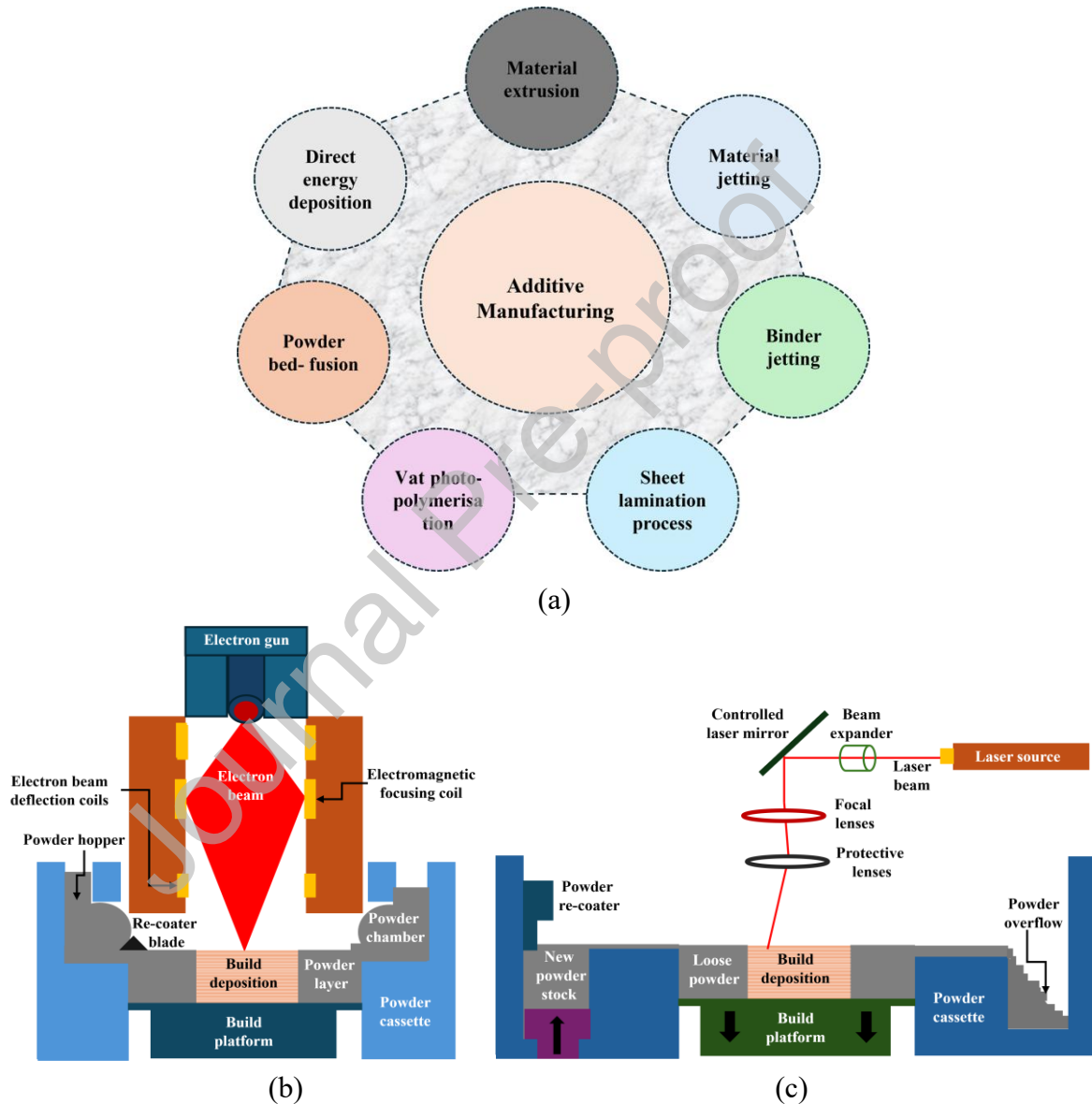


Fig. 1. (a) ASTM lists different types of additive manufacturing processes, (b) Block and line diagram of the EB-PBF system, (c) Block and line diagram L-PBF system [9,11].

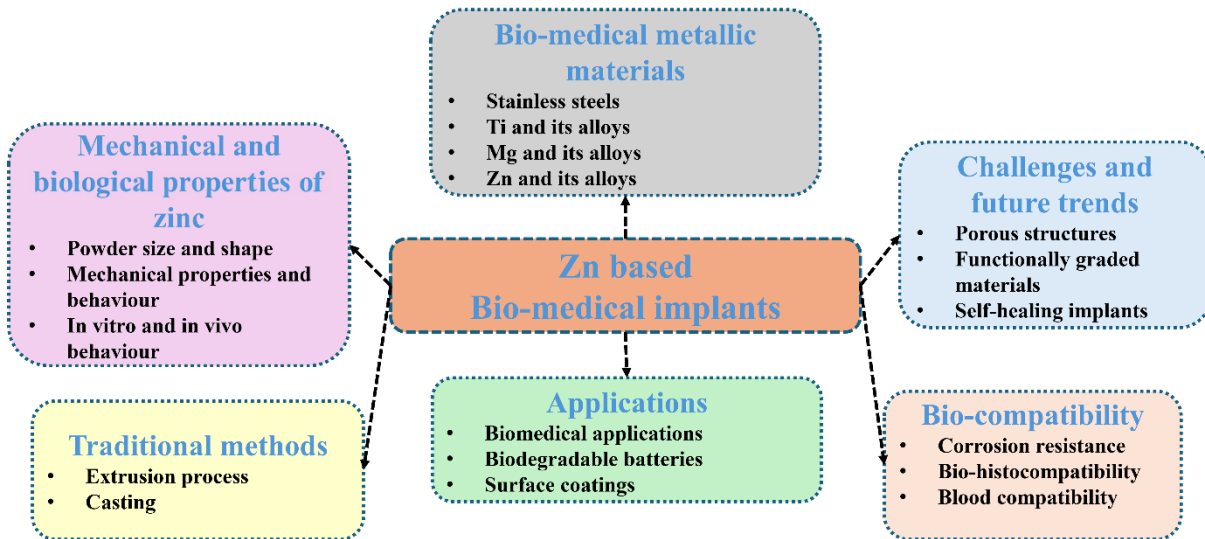


Fig. 2. Scope of the review.

1.1 Metal implants and their needs

When original human parts are worn out, injured, or incapable of independent functioning, implants can alleviate issues with the bones and joints within the human body. They may need replacement for various reasons, including osteoarthritis, scoliosis, spinal stenosis, bone fractures, and persistent discomfort. For instance, during the self-healing phase, broken bones are stabilized with pins, rods, screws, and plates [36]. Conventional implants consist of iron, copper, stainless steel, titanium, magnesium, and their alloys. These materials have been largely abandoned due to minute disintegration, reactions within the body, and the release of toxic ions. Since the 1950s, most implants have been constructed from stainless steel (SS) and titanium (Ti) based alloys. Their corrosion resistance, mechanical properties, and low cost facilitate the production of cranial implants, spinal rods, dental implants, joint replacements, and orthopaedic fracture plates, while stabilising biological tissues and structures [37]. In orthopaedic surgery, a major challenge is treating significant bony defects, which are limited by the availability of suitable substitute materials for ideal bone [38,39]. Besides biocompatibility, an ideal metal or alloy substitute must closely match mechanical properties to provide mechanical strength, avoid stress shielding, support bone ingrowth through fully interconnected porous structures, and degrade at the same rate as bone regeneration [40,41]. Recent advances in AM can address these challenges by incorporating bone-mimicking properties and interconnected porous structures, utilising optimal AM process parameters [42,43]. Bio-inert materials can form interconnected pores with bone-mimicking properties, including tantalum, chromium-cobalt, titanium alloy, and stainless steel [44]. However, these implants do not degrade over time, are unsuitable for bone regeneration, and may cause infections due to bacterial colonisation of their surfaces [45]. They may also result in chronic inflammatory reactions, permanent physical irritation, and long-term endothelial dysfunction [46]. Biomaterials that satisfy these limitations and are biodegradable are suitable for the orthopaedic and traumatological communities.

1.2 Causes of permanent implant failures

Reasons for implant failure include pain, wear, infection, bio-incompatibility, aseptic loosening, and the need for revision surgery. The most common causes are described here.

(1) Wear of implant material

Wear is the most critical challenge for implants, particularly at joints such as the elbow, knee, hip, and medial proximal tibia. It may result in the loss of bulk in implanted materials, seriously impairing the functionality of orthopaedic prostheses. For example, osteolysis in total hip arthroplasty (THA) may result from wear debris from the polyethylene acetabular components, as shown in Fig. 3(a) [47,48]. Similarly, vanadium (V) in a titanium alloy such as Ti-6Al-4V has been found to harm human health and accumulate in organs. Ion poisoning results from the leakage of V ions into the in vivo environment when the stable coating, TiO_2 , wears down [49,50].

(2) Bacterial infection of implants

The skin surrounding the incision made during implant placement and the doctor's hands are primary sources of germs causing implant infections. During hematogenesis, germs migrate from inside the patient's body through incision channels to the surface of the implants. The initial and most crucial stage in bacterial infections of implants is bacterial adherence. Bacteria cling to the implant and become stationary, providing sites for biofilm formation and for the attachment and multiplication of bacteria, as shown in Fig. 3(b). Adherent bacteria on the surfaces of implanted medical devices produce an extracellular polymeric material that shields the germs from elimination. In its absence, bacteria may move freely throughout the body [51].

(3) Loosening of implants

In approximately 32.6% of orthopaedic cases, implant loosening has been identified as the most common cause of failure. A contributing factor to implant loosening is stress shielding. This biological phenomenon results in osteopenia, in which bone mineral density is lower than normal because the implant absorbs forces previously borne by the bone itself. Wolff's law states that the bone of a healthy human or animal will remodel in response to typical loads, thereby influencing bone density. Strength or density diminishes when stress or load is reduced, as no stimulus is required for bone remodeling to sustain bone density. Similar issues are prevalent in dental contexts, as shown in Fig. 3(c) [52,53].

(4) Corrosion of the implant alloy

The most common and easily visible types of corrosion in clinical settings are crevice, galvanic, and pitting corrosion. It should be emphasized that many types of corrosion may occur independently or concurrently and can complement or prolong previously observed corrosion. The intricacy of the electrochemical contacts between the superstructure and the implant stems from pitting corrosion induced by galvanic coupling. Galvanic corrosion results from ion exchange between implants and their prosthetic components. Pitting corrosion is often observed at the implant-abutment interface, as shown in Fig. 3(d) [54]. Implants corrode upon contact with bodily fluids after implantation. The body's environment is corrosive due to

proteins and chloride ions, which induce various chemical reactions on the implant surface. Dissolved oxygen is reduced to hydroxide ions, and the alloy's metallic components are oxidized to their ionic forms [55].

(5) Revision or secondary surgery

In orthopaedics and cardiovascular surgery, secondary surgery is needed to remove permanent implants, as they increase the risk of long-term infection [56]. Secondary surgery may also be required after bone healing [57]. These procedures are more painful and necessitate a new recovery period for patients, imposing an overall financial burden on both the patient and the healthcare system. Chronic inflammation, physical irritation, and vascular abnormalities are the most widely recognized disadvantages of permanent implants [58]. Paediatric challenges are critical, as infant bodies are growing, and permanent implants cannot accommodate this natural growth [59,60].

Biodegradable metallic implants are now used instead of permanent metallic implants due to the limitations outlined above. Biodegradable implants are materials that degrade or are gradually reabsorbed, allowing new cells to form functional tissue. When placed at the disease site, they reduce the risk of stress shielding and implant loosening. These materials provide initial mechanical support to fatigued organs or tissues and are progressively resorbed by biological fluids, promoting bone tissue regeneration in critical-sized bone defect areas. They do not harm human health after wear, as their corrosion products are uniform, biocompatible, and non-toxic [61].

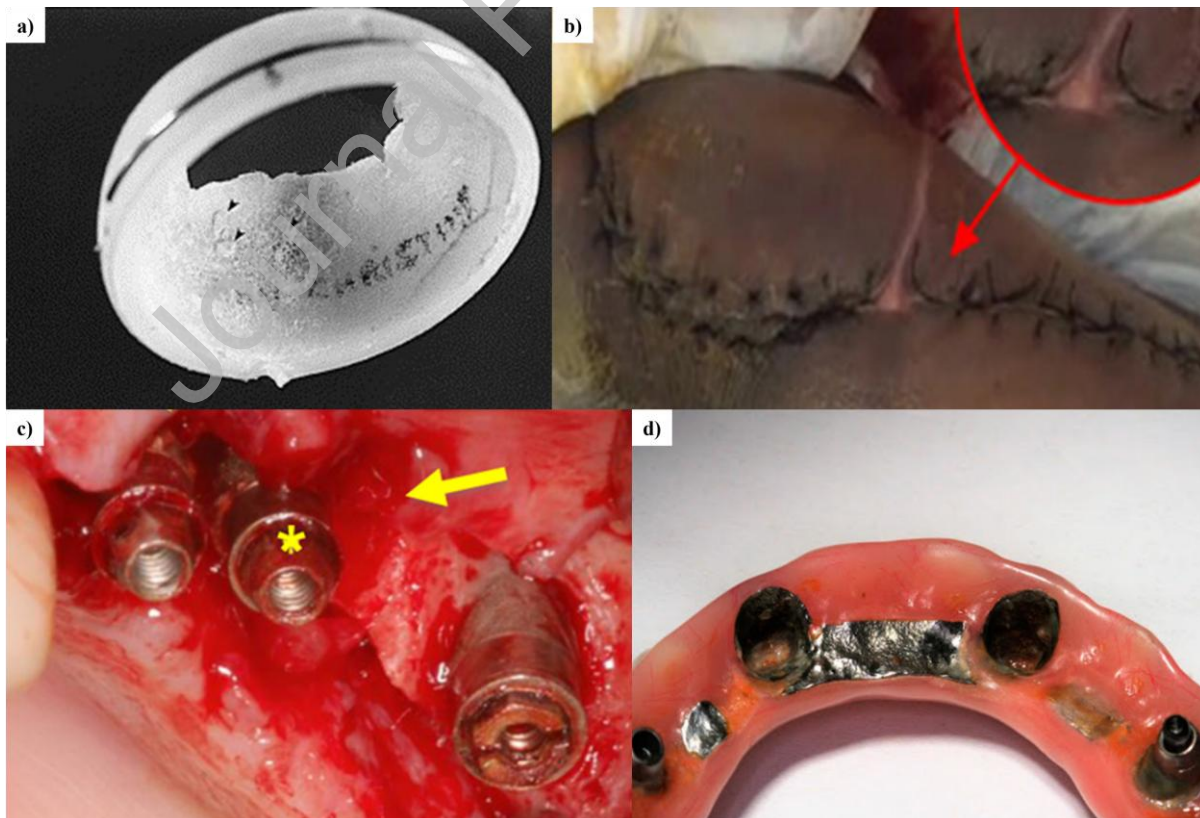


Fig. 3. (a)Acetabular implant exhibits aseptic loosening attributable to wear ⁴⁸, (b)

contamination leading to the progressive onset of infected osteolysis and fistula ⁶², (c) Bone resorption around the implant resulting in implant loosening ⁵³, (d) Corroded superstructure hybrid prosthesis ⁵⁴.

1.3 Biodegradable metals (BMs) and their applications in the medical domain

AM in the medical sector and allied fields (orthopaedics, dental implants) has several biological, technological, and economic objectives, including biocompatibility, cytocompatibility, functional performance, strength, lightweight design, reduced lead time, cost containment, sustainability, and degradation time ⁶³. Temporary metal implants are biodegradable and absorbed by the body, unlike standard permanent implants such as stainless steel and titanium ⁶⁴. Metal implants provide greater strength, mechanical properties, and degradation characteristics than polymer biodegradable implants such as polylactic acid (PLA) ^{65,66}. Biodegradable metal implants (BMI), including bone fracture fixation devices and cardiovascular stents, are extensively researched and widely available in the market ^{67,68}. Bioabsorbable metals and their alloys corrode slowly in vivo, releasing corrosion products that can be metabolised and assimilated by cells and tissue before complete dissolution, promoting tissue healing and eliminating implant residues ⁶⁹. BMs are classified as absorbable metallic materials according to ASTM F3160, and the terms absorbable, bioabsorbable, and bioresorbable are used interchangeably to denote the same concept ⁷⁰. Highly rigid and strong bone plates stabilize fractures during surgical procedures due to much higher stiffness compared to natural bone. However, this mismatch causes prolonged mechanical load bearing by the bone. As healing progresses, the bone receives insufficient mechanical cues or loads in the later stages, potentially leading to complications such as osteoporosis. The phenomenon termed the "stress shielding effect" arises from the refracture of bone following implant removal from the healing site ⁷¹. Biodegradable implants are preferred for two key benefits: they gradually shift mechanical stress to regenerating tissue and eliminate the need for additional procedures to extract the device. These implants break down within the body after fulfilling their role, ultimately supporting patient recovery without long-term intervention ^{69,72}. There is a close connection between these goals, making the selection of the best design solution contingent on careful consideration of all relevant factors. The significance of these objectives varies based on the medical application. The medical implant industry demands small-scale production quantities of safety-critical parts with strict delivery schedules that must function effectively in their intended environment. Therefore, lead-time reduction is crucial, as it enables rapid product certification and preserves design freedom for high-value components. Generally, implant structures are biocompatible, low-volume, high-complexity systems, where significant sustainability and degradation issues may arise. Challenges associated with sustainability include remanufacturing and repair, recertification of existing implants for other applications, and ensuring part availability for aging implants. Intricate and porous support structures were fabricated using L-PBF, employing various degradable metal blends. Some alloys contained rare-earth elements, which were observed to accumulate in organs such as the

liver, kidneys, spleen, and lungs during animal studies involving sheep, rats, and rabbits. Consideration must be given to alternative elemental compositions to prevent such accumulation^{63,73}. Bioabsorbable metallic compounds, including Iron (Fe), Magnesium (Mg), and Zinc (Zn), have attracted significant scientific interest worldwide due to their potential applications in orthopaedics⁷⁴. Iron and magnesium, along with their alloys, have been recognized as promising biodegradable metallic materials, particularly in cardiovascular and orthopaedic applications. Both elements are vital micronutrients, with recommended daily intake ranging from 240 mg to 420 mg for magnesium and 8 mg to 18 mg for iron, depending on age and gender⁷⁵. Magnesium is essential for various bodily functions, including cellular growth, energy usage, internal signaling, and molecular movement within cells. It exhibits excellent compatibility with biological systems and poses a low risk of triggering blood clot formation⁷⁶. Mg is particularly suitable for biological applications due to its mechanical behavior resembling natural bone⁶⁴ and is crucial for bone growth, human nutrition, and bone formation⁷⁷. Clinical experiments have been conducted on degradable bone screws, stents, and plates⁶⁸ composed of magnesium-based alloys, which became commercially accessible in 2016⁷⁸. However, while magnesium provides certain benefits, its rapid degradation in bodily environments presents a significant challenge. This swift breakdown leads to the accumulation of hydrogen bubbles around the implant, which can compromise mechanical stability during the recovery period⁷⁹. Certain limitations are associated with the use of Magnesium and its alloy-based implants, including rapid deterioration, hydrogen (H) production, and heterogeneous or inhomogeneous corrosion^{80,81}. The corrosion of magnesium releases hydrogen gas, necessitating careful control of the corrosion rate, a challenge in many medical applications⁸². Mg alloys are also susceptible to fatigue corrosion (FC) and stress corrosion cracking (SCC). Residual stress, likely present during manufacturing or implantation, induces FC and SCC, accelerating degradation and the attenuation of mechanical strength⁸³. To manage both Mg corrosion rate and mechanical strength, alloying elements such as rare-earth elements are incorporated^{79,84,85}. Iron (Fe), being bioavailable, non-toxic, and essential for deoxyribonucleic acid (DNA) synthesis and the production of oxygen-rich molecules such as haemoglobin and myoglobin, exhibits a low corrosion rate, leading to a disparity in the body's absorption rate. Elevated mismatch modulus induces stress shielding. Zn-based bioabsorbable metallic alloys offer an opportunity to bypass these limitations and have recently attracted the attention of several researchers. According to the reviewed paper, a notable increase in publications on pure zinc occurred in 2022, with China leading in the number of papers, as shown in Fig. 4(a) and Fig. 4(b). where pure Zn has garnered has attracted significant interest as a biodegradable metal due to its biocompatibility, non-toxicity, and corrosion resistance. The biodegradable properties of the metal designated for implantation are discussed in Table 1. Based on the manufacturing process, metals are produced via L-PBF, enhancing the density, shape, and microstructure of Zn components, which affects their mechanical characteristics and deterioration rate. The quality of standard implants, such as screws, rods, and cardiovascular clips, may be improved by employing the L-PBF process^{86,87}. Zinc is generally

considered a self-corrosion element, also referred to as electrochemical corrosion. Zinc-based implants exhibit a standard hydrogen electrode (SHE) corrosion potential of -0.76 V (vs SHE), making Zn more noble than magnesium (-2.37 V vs SHE) but less noble than iron (-0.44 V vs SHE). This intermediate potential results in a modest deterioration rate of $14\text{--}30$ $\mu\text{m}/\text{year}$, corresponding with the normal timetable of bone repair (3–6 months)⁶⁹. Conversely, Zn and Fe-based alloys do not produce hydrogen during corrosion and provide excellent resistance to stress corrosion cracking and corrosion fatigue, thereby circumventing challenges associated with Mg-based implants⁸³. From this perspective, zinc and its alloys show significant promise for use in small-scale implants, such as dental fixtures, orthopedic implants, and cardiovascular stents. The modulus of elasticity and mechanical strength of pure zinc, produced by different manufacturing processes, are presented in Table 2. However, the appropriateness of zinc for larger implants warrants further examination due to potential toxicity associated with elevated zinc release^{88,89}. Due to zinc's relatively low melting and vaporization points, the L-PBF process can result in low-density components. Process parameters must be precisely controlled to produce high-density parts and fully leverage the benefits of L-PBF for biodegradable Zn alloys. In L-PBF, the applied energy density (fluence) significantly impacts porosity. Insufficient energy input may cause porosity due to incomplete melting, while excessive fluence may disturb the melt pool^{90,91}. The effectiveness of forming compact parts depends heavily on the material type. Achieving structurally sound components with reliable mechanical performance necessitates steady energy input throughout fabrication to ensure consistent density. Over time, as smoke and particulate matter accumulate in the chamber, the laser beam may lose intensity or be redirected by interference. Zn possesses relatively low thresholds for melting and vaporization. Key operational factors in the process include beam intensity (P), movement rate or scanning speed (v), the distance between successive paths and tracks or layer hatching (h), and the depth of each deposited layer or layer thickness (t)⁵⁹. Table 1 summarizes the characteristics required for bioabsorbable metals used in orthopaedic implants [64,92,93]. Table 2 list the modulus of elasticity and mechanical strength of pure zinc (Zn) tensile specimens by different manufacturing processes [33, 60, 93-101].

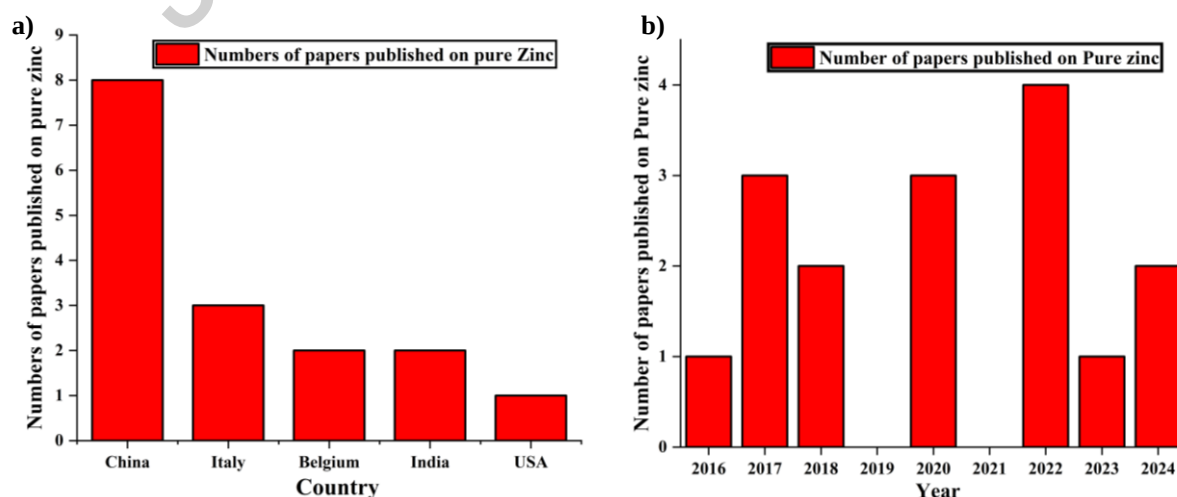


Fig. 4. Number of Papers published in pure zinc: (a) Country-wise from (2016-2024); (b) Year-wise.

Table 1. Characteristics of the bioabsorbable metals required for orthopaedic implants ^{64,92,93}.

S. No	Characteristics of the metals	Magnesium and its alloys	Zinc and its alloys	Iron and its alloys
1.	Essential elements involved in the regulation of skeletal tissue processes	Yes	Yes	Yes
2.	Advised routine intake for optimal regular use	0.375-0.500 g	0.002-0.015 g	0.006-0.020 g
3.	The usual amount of food and nutrients consumed per day	0.329 g	0.0086 g	0.0132 g
4.	The level of specific substances in the liquid portion of the blood is expressed in millimolar (mM)	0.73-1.06 mM	0.012-0.017 mM	0.013-0.031 mM
5.	Poisson coefficient.	0.35	0.30	0.27
6.	Ultimate tensile strength (UTS) MPa	199-350	167-520	567-1059
7.	Young modulus MPa	41000-45000	94000-110000	204000-212000
8.	Standard reduction potential (vs. SCE)	-2.37V	-0.76V	-0.44V
9.	In vitro rate of deterioration or the degradation rate	160.0-1660.0 $\mu\text{m/yr}$	450.0-12560.0 $\mu\text{m/yr}$	40.0-960.0 $\mu\text{m/yr}$
10.	Influence of surrounding acidity levels throughout the breakdown process	Alkaline	Alkaline	Alkaline
11.	Corrosion mode type	Mostly localised corrosion and pitting	Uniform or pitting corrosion	Localised corrosion
12.	Primary corrosion byproducts	Mg(OH) ₂ , MgO, MgCl ₂	Zn(OH) ₂ , ZnO, ZnCO ₃	Fe(OH) ₂ , Fe ₃ O ₄
13.	Emissions generated through the disintegration process	Hydrogen	None	None
14.	Positive influences on skeletal tissue health	Crucial for developing a healthy skeletal system; it helps guard against bone thinning, persistent joint calcification, and weakened bone structure.	Vital for new bone formation, it prevents bone loss by reducing bone-building cells and increasing bone-resorbing cells, helping maintain strong bones.	Promotion of bone growth

Table 2. Modulus of elasticity and mechanical strength of pure zinc (Zn) tensile specimens by different manufacturing processes.

S. No.	Manufacturing process	Young's modulus (GPa)	Ultimate tensile strength (MPa)	Yield strength (MPa)	Elongation at break (%)	Ref. No.
1.	Casting	-	52.7 $\pm$ 4.6	47 $\pm$ 4.2	0.31 $\pm$ 0.02	[33]
2.	Casting	-	18.25	10.14	0.32	[94]
3.	Casting	-	29.75	22.85	3.34	[95]
4.	Casting	-	33.6	29.3	1.2	[96]
5.	Hot-rolled	-	50	30	6	[94]

6.	Extrusion	-	100	55	7.5	[97]
7.	Extrusion	-	60	35	3.5	[98]
8.	Extrusion	-	180	130	53	[94]
9.	L-PBF	-	127.8	100.7	7.6	[99]
10.	L-PBF	12±2.4	61 ± 5.0	43±3.1	2 ± 0.1	[100]
11.	L-PBF	20 ± 5.7	138 ± 2.5	122 ± 2.6	8 ± 0.6	[96]
12.	L-PBF	36	113	102	11	[101]
13.	L-PBF	-	123.6 ± 8.2	109.5 ± 7.6	-	[33]
14.	L-PBF	53-88 ±1	79-100 ± 0.5	55-78 ± 0.5	10-12 ±0.5	[63]

2. Additive manufacturing of Zinc and its alloys

2.1 Factors affecting the build quality

2.1.1 Powder Characteristics

The main need of the L-PBF component is superior quality and enhanced surface polish. The quality of parts produced in L-PBF depends on the type of powder used in the process. Optimal criteria for selecting powder for the L-PBF process include high purity, relative density, uniform particle size distribution, consistent composition, and excellent sphericity. Three methods of powder atomisation are recognized in the L-PBF process: gas, air, and water atomisation, as shown in Fig. 5 [96, 102 -105]. The shape of the powder indicates the processing conditions: the nitrogen-atomised powder is spherical, while air- and water-atomised powders are irregular in shape and size [102]. The zinc powder particle sizes reported by the authors are shown in Fig. 6 [16,18,56,57,59,96,103,106-110] .

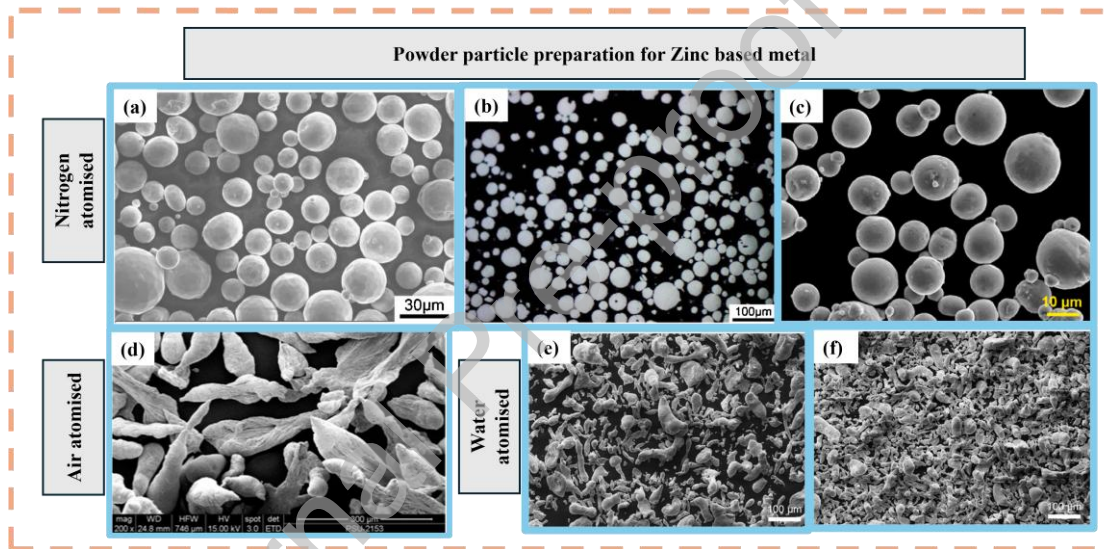


Fig. 5. Microstructure of pure zinc powder particles by SEM: (a), (b), and (c) are developed by Nitrogen atomisation [96,103]; (d) by air atomisation [104]; (e) and (f) are by water atomisation [105].

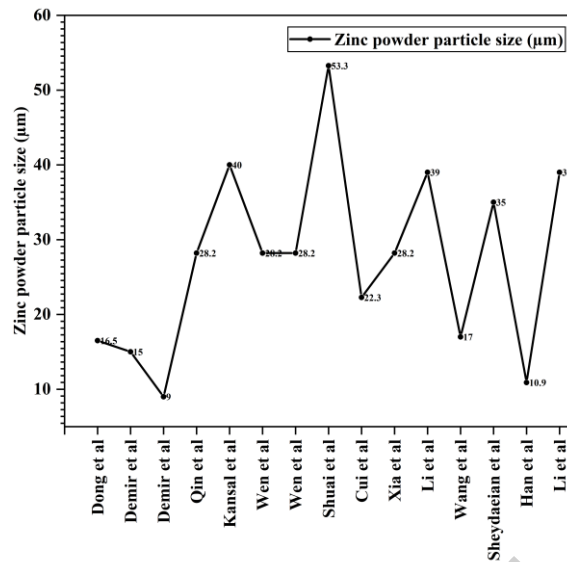


Fig. 6. Zn powder average particle size across different experiments [16,18,56,57,59,96,103,106-110].

2.1.2 Laser processing parameters and the environment

Components manufactured using the L-PBF process may contain defects, such as pores and cracks, due to complex physical and chemical interactions, e.g., laser concentration, laser reflection, material vaporisation, and heat transfer, which can significantly influence overall performance [111,112]. The L-PBF of Zn metal, conducted within a sealed processing chamber, quickly saturates the surrounding atmosphere. During L-PBF, excessive vaporisation and particle ejection may compromise the quality of the printed component due to rapid chamber contamination. Metallic vapours and particles in the surroundings may disrupt laser beam transmission and compromise optical components, including the protective glass of the scanning head. This degradation may occur gradually as the chamber becomes more contaminated, particularly during extended processing of large or intricate components. Fig. 7(a) shows that using a processing or inert gas to treat zinc in an enclosed environment quickly contaminates the chamber, leading to dispersion of the laser beam and insufficient material melting. The low vaporisation temperature of pure Zn leads to substantial vapour production during layer development, causing material expulsion from the processed or focused region. Under steady-state processing conditions, the processable range includes partial fusion at low energy densities (below 40 J/mm³). Conversely, higher material density is achieved at moderate energy levels when using coarse powder, as shown in Fig. 7(b). Although stability is maintained for a few initial layers, higher energy densities (above 115 J/mm³) may lead to the disintegration of the part, causing previously deposited layers to burst out from the build. Additionally, fine powders are more susceptible to vaporisation at the same energy input and should be processed under lower fluence levels, as depicted in Fig. 7(c). A clean, controlled atmosphere must be sustained in the process chamber to ensure optimal conditions. If this is not feasible, operating with an open chamber may serve as an alternative to prevent atmosphere saturation⁵⁹. Similarly, flaws may appear during layer-by-layer construction, impairing the part's mechanical

and physical properties or geometric precision. Investigations were conducted to determine whether plume features may signify process stability in L-PBF technology. The primary notion posits that, under regulated settings, the statistical characteristics of the plume should remain stable over time. Consequently, the first layers of the construct may be utilized to assess the process's dynamic behaviour through quantifiable indicators, which can guide control strategies for subsequent build layers. For zinc, characterized by a low melting point, evaporation significantly influences part quality during laser powder bed fusion (L-PBF). This evaporation produces strong odours and results in splatter and spatter. The resultant fumes contain numerous minute particles that obstruct the transmission of the laser beam in subsequent melting phases. The attenuation of laser energy destabilizes the melting process, eventually undermining the quality of the produced component [113].

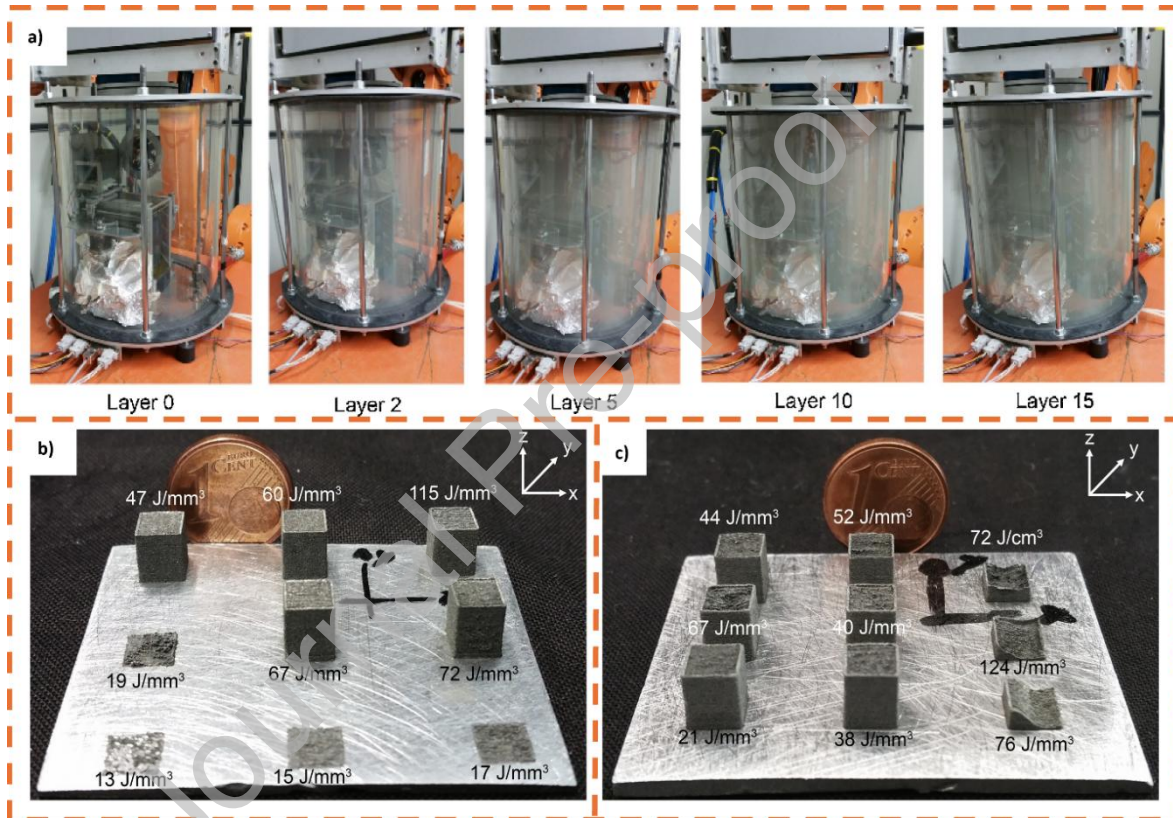


Fig. 7. (a) Processing chamber accumulates Zn vapour and particles as the number of deposited layers builds; (b) Inadequate fluence combined with coarse powder results in incomplete melting and leads to missing sections of the part; (c) Excessive fluence combined with fine powder causes an explosive reaction, resulting in part loss [59].

The gas flow system in the L-PBF chamber has been enhanced with a gas circulation system, designed through numerical modeling, to minimize the evaporation of zinc metal into the environment. Significant evaporation results in a substantial amount of melted Zn being expelled from the molten pool. Excessive hydrodynamic forces push surrounding particles out of the molten pool, disrupting the ejection process. Hardened spatters create and partially melt circular or acicular particles that attach to the surface, increasing roughness and diminishing

surface finish. The volume of evaporation for a particular substance is dictated by the energy of the laser; increased input of laser energy results in elevated temperatures and enhanced evaporation rates. The volume laser energy density (Ev) is a key metric for quantifying the energy input of the laser beam throughout the L-PBF process. In a similar manner, the laser power (P), scanning velocity (V), and hatch spacing (H_s) are critical parameters for assessing the stability of the process, as defined in Eq. (1) and also shown in Fig. 8(a) ⁹⁶.

$$Ev = \frac{P}{V \times H_s \times D_s} \text{ J/mm}^3 \quad (1)$$

The ideal relative density of the deposited material in the sample should approach 100%; however, achieving this in experiments is impractical. According to Wen et al., the reported relative density of pure zinc in the L-PBF process is 99.5% [96]. Other authors observed that the relative density initially increased, then declined as Ev increased. At a low Ev of 23.8 J/mm³, the density dropped to 93.33%, with numerous irregular pores visible in the cross-sectional microstructure. This was attributed to insufficient laser energy, resulting in incomplete fusion between layers. At low Ev values, inadequate wetting of the molten metal led to void formation when the liquid metal failed to fill gaps during solidification. An increase in Ev to 85.7 J/mm³ resulted in a notable density enhancement to 99.91%, attributed to effective melting and improved wetting behavior. At Ev values of 190.5 J/mm³, the density dropped to 97.88%, accompanied by increased porosity. The pores appeared more spherical and exhibited random distribution under higher energy inputs. The principal sources of these problems were the trapping of zinc vapor or shielding gas within the molten pool. High levels of laser energy caused significant metal evaporation. The generated vapor plume could exceed the shielding gas's capacity to completely evacuate it, leading to vapor condensing into fine particles that collected within the build chamber. The presence of these particles disrupted laser transmission, resulting in variable energy absorption across the powder bed and increased porosity. Additionally, vigorous evaporation disrupted the molten pool, fostering turbulence and increasing the likelihood of gas entrapment, further promoting pore formation [114]. Similarly, other metals alloyed with zinc exhibited porosity in the powder bed process due to increased energy input [115]. The optimal volumetric energy density of pure zinc is approximately 127 J/m³, much lower than that of titanium metal at 289 J/m³ in the L-PBF process, attributable to its reduced melting and boiling temperatures, as shown in Fig. 8(b) [116]. In a similar experiment, the optimized process parameters for pure zinc ranged from 60 J/mm³ to 135 J/mm³, resulting in material deposition with a high relative density of 90–99%, as illustrated in Fig. 8(c) [56]. Cui et al. outlined the correlation between laser power and scanning speed to elucidate the connection of process parameters for pure zinc, delineating defects based on relative density and volumetric energy density. Desired densification behavior was attained, as shown in Fig. 8(d) [56,108]. The processing parameters are the main pillars of the L-PBF process that confer unique microstructural characteristics and relate to the bulk properties of Zn and its alloys; these parameters are presented in Table 4.

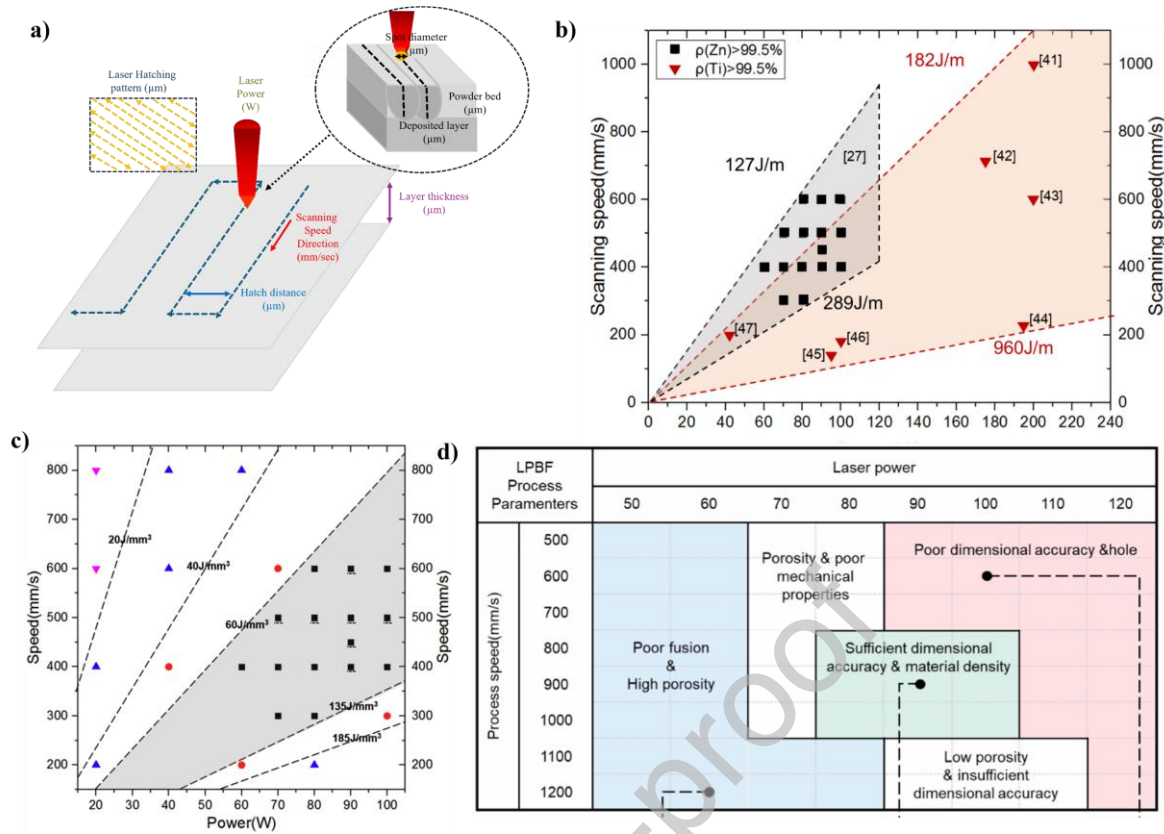


Fig. 8. (a) A line diagram illustrating the L-PBF process, (b) Parameters of the L-PBF process for Zinc and Titanium¹¹⁶, (c) The connection between volumetric energy density, scanning speed, and laser power⁵⁶, and (d) Mapping the laser power and scanning speed for pure zinc¹⁰⁸.

2.1.3 Microstructure

The grain size in AM is determined by the cooling rate during solidification, typically ranging from 10^6 to 10^5 K/s. A higher cooling rate results in a finer microstructure, enhancing the material's strength [117,118]. The finer grains produced by AM techniques result in relative densities exceeding 88% for L-PBF, conferring superior strength compared to conventional production processes⁶⁰. The Archimedes method was used by Yang et al. to determine the relative density of the zinc L-PBF-processed component. The zinc microstructure is dense and free of holes or fissures, with a relative density of $97.4\% \pm 1.8\%$ ¹⁰⁰. Metallographic analysis during zinc deposition indicated porosity up to 1.54%. The deposited material exhibited pores with mean and median diameters of $5.8 \mu\text{m}$ and $3.7 \mu\text{m}$, respectively. The observed pore sizes suggest that the energy input was sufficiently high to eliminate lack-of-fusion defects while also carefully regulated to prevent keyhole-induced porosity and gas entrapment⁶³. Cross-sectional microstructures, as noted by Qin et al.¹⁰⁶, were aligned perpendicular to the normal direction (ND). The absence of a discernible interface between layers indicates strong metallurgical bonding in deposited Zn layers. Several twin crystals were identified within the columnar grains, which constituted the majority of the microstructure and aligned with the building direction (BD). The granules became finer with an increase in scanning speed, as shown in Fig. 9(a)¹⁰⁶. Lack of fusion flaws attributed to inadequate energy density partially

caused the zinc powder to melt, as demonstrated by Mahdi et al. in Fig. 9(b). Samples fabricated with low energy densities (40–60 J/mm³) often exhibit this defect. Conversely, similar flaws can arise at excessively high energy densities due to intense evaporation during processing, commonly observed in samples with energy inputs exceeding 100 J/mm³. Zinc processed at an optimal energy density range (70–80 J/mm³) showed a more refined microstructure. In this range, keyhole-induced defects were the most prevalent, while issues associated with excessive evaporation or incomplete melting were minimal. The use of triply periodic minimal surface (TPMS) designed cell structures is beneficial in biomedical applications. Scanning electron microscopy (SEM) photographs in Fig. 9(c) depict various defects in the fabricated specimens, primarily micropores, and variations in the wall thickness of the scaffolds. These defects arose during chemical cleaning or resulted from evaporation during the L-PBF process due to the low melting and boiling points of zinc. Processing of Zn by L-PBF encounters significant challenges due to substantial evaporation, leading to porosity and reduced density in the produced components. Although the impacts of evaporation were mitigated by modifications to the inert gas circulation system and improvements in process parameters, some flaws remained in the finished scaffolds [119].

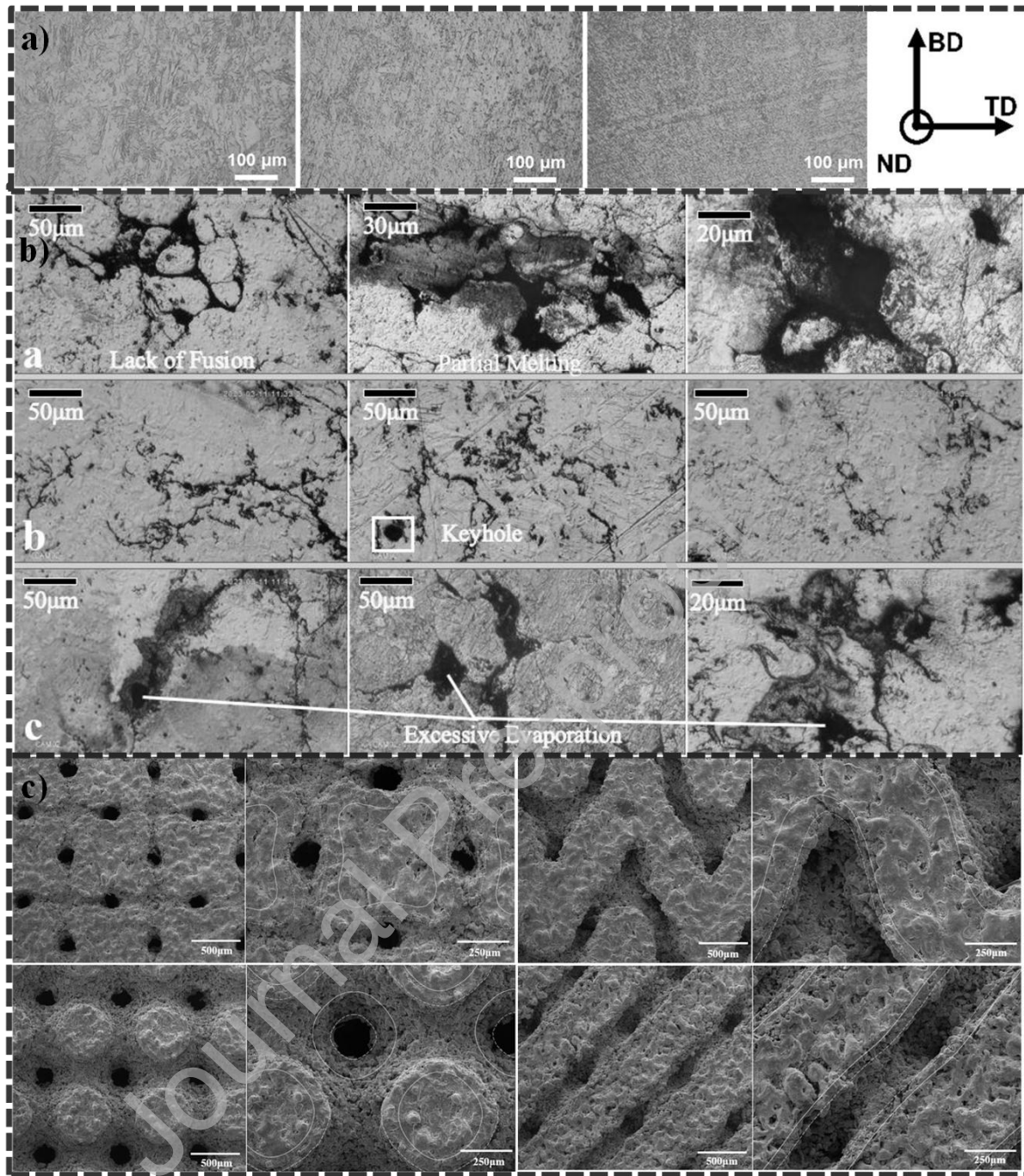


Fig. 9. (a) Microstructure of LPBF pure Zn samples deposited at 300, 500, and 700 mm/s scanning rates; (b) SEM images of cubic samples fabricated under varying energy density; (c) SEM micrographs depicting the morphology of individual TPMS ^{106,119}

Qin et al. ¹⁰⁶ reported that horizontal samples exhibit mechanical anisotropy, demonstrating reduced strength and ductility compared to vertical samples. According to Dong et al., microcracks initiate at the start and propagate along grain boundaries during tensile testing ³³. Zinc displays promising build quality at optimised laser energy density and shielding gas. Qin et al. indicated that the relative density of the deposited sample exceeds 99.5%. XRD patterns exclusively show peaks corresponding to the hexagonal close-packed (hcp) crystal structure,

with no evidence of secondary phases or peaks associated with zinc oxides. The microstructure consists mainly of columnar grains along the building direction, exhibiting excellent bonding between layers and no visible interface. Grain orientation maps reveal finer grains at higher scanning speeds and in orientations of the $\langle 0001 \rangle$ plane due to twinning, primarily occurring at the $\langle 1012 \rangle$ plane in the hexagonal crystal, where the lattice parameters are a , b , and c . The c/a ratio of zinc approaches the ideal hexagonal crystal value of 1.732 [120]. Finer grains are observed when scanning speed exceeds 500 mm/sec. Additionally, texture influences zinc characteristics, as it is affected by the temperature gradient during the solidification step of the L-PBF process. A firm texture leads to a loss of ductility and increased anisotropy. Pole Figures for the crystallographic planes $\{0001\}$ and $\{1010\}$ are provided. In the sample analysed at a scanning velocity of 300 mm/s, $\{0001\}$ pole Figures reveal a pronounced $\langle 0001 \rangle$ crystallographic texture aligned with the normal direction, diminishing progressively with increasing scanning speed. The $\{1010\}$ pole Figures indicate that the $\langle 1010 \rangle$ texture is detected only along the build direction in the 300 mm/s sample. Maximum pole Figure intensities at scanning speeds of 300, 500, and 700 mm/s were documented as 36.618, 14.584, and 8.081, respectively, as shown in Figs. 10(a–c) [106]. Consequently, LPBF Zn samples display deteriorating grain texture with increasing scanning speed [106,121,122]. Karel et al. show that Dense L-PBF Zn exhibits substandard crystallographic texture, with minimal influence from hatch pattern rotation between layers on its intensity. The rotation of hatch patterns across layers influences the orientation of grain structure within the build plane. The $\langle 0001 \rangle$ crystallographic direction predominantly aligns with the build direction. In the absence of hatch pattern rotation between consecutive layers, the $\langle 1010 \rangle$ direction primarily aligns with the build direction, whereas the $\langle 0001 \rangle$ direction is oriented along the scanning direction (X -axis), as shown in Fig. 10(d). Rotating the hatch pattern by 90 degrees between successive layers affects grain orientation, while Fig. 10(e) illustrates that the $\langle 0001 \rangle$ crystallographic direction is predominantly oriented at a 45-degree angle relative to the scanning directions (X and Y). The $\{0001\}$ pole Figure exhibits asymmetry, potentially due to minor polishing and grinding defects or misalignment of the measuring plane. Fig. 10(f) demonstrates that the $\{0001\}$ crystallographic direction is predominantly aligned within the build plane, featuring a nearly random distribution within this plane due to layer-by-layer variation in hatch pattern orientation [63,94].

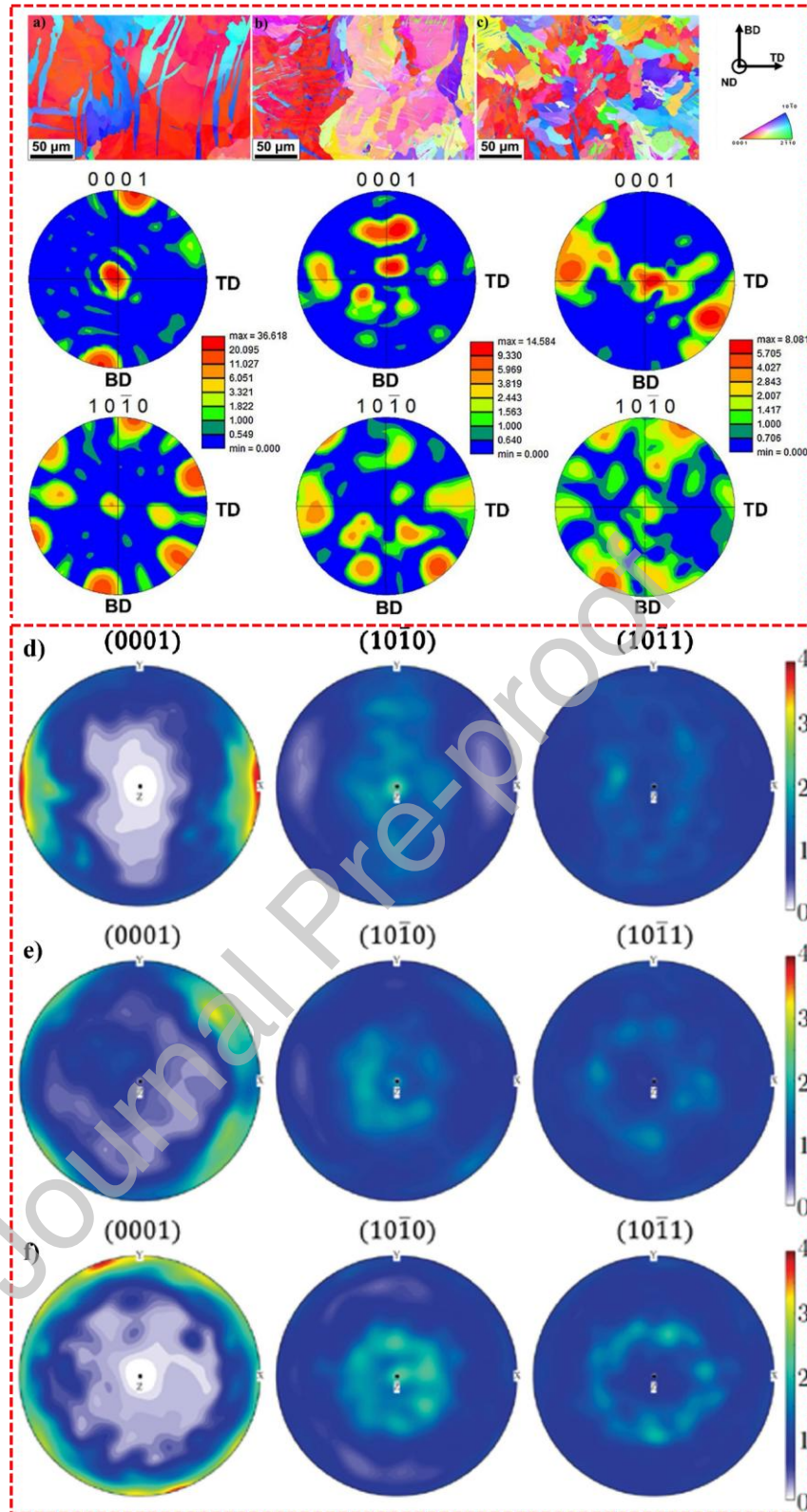


Fig. 10. The crystalline structure of the dense LPBF Zn and the EBSD orientation maps of LPBF-fabricated pure Zn samples processed at 300 mm/s, 500 mm/s, and 700 mm/s, respectively, are shown in (a), (b), and (c). The construction direction was mostly aligned with the 0001 crystallographic direction; The in-plane orientation was affected by the hatch pattern rotation between layers: (d) 0°, (e) 90°, and (f) random rotation^{63,106}.

2.1.4 Mechanical properties

The mechanical characteristics of Zn-based biomedical implants are shown in Fig. 11 [20,56,60,63,96,100,103,106,123-138]. Excellent mechanical properties have been exhibited, and effective production has been achieved using the L-PBF process in recent years. These properties are superior to those obtained through conventional processes.

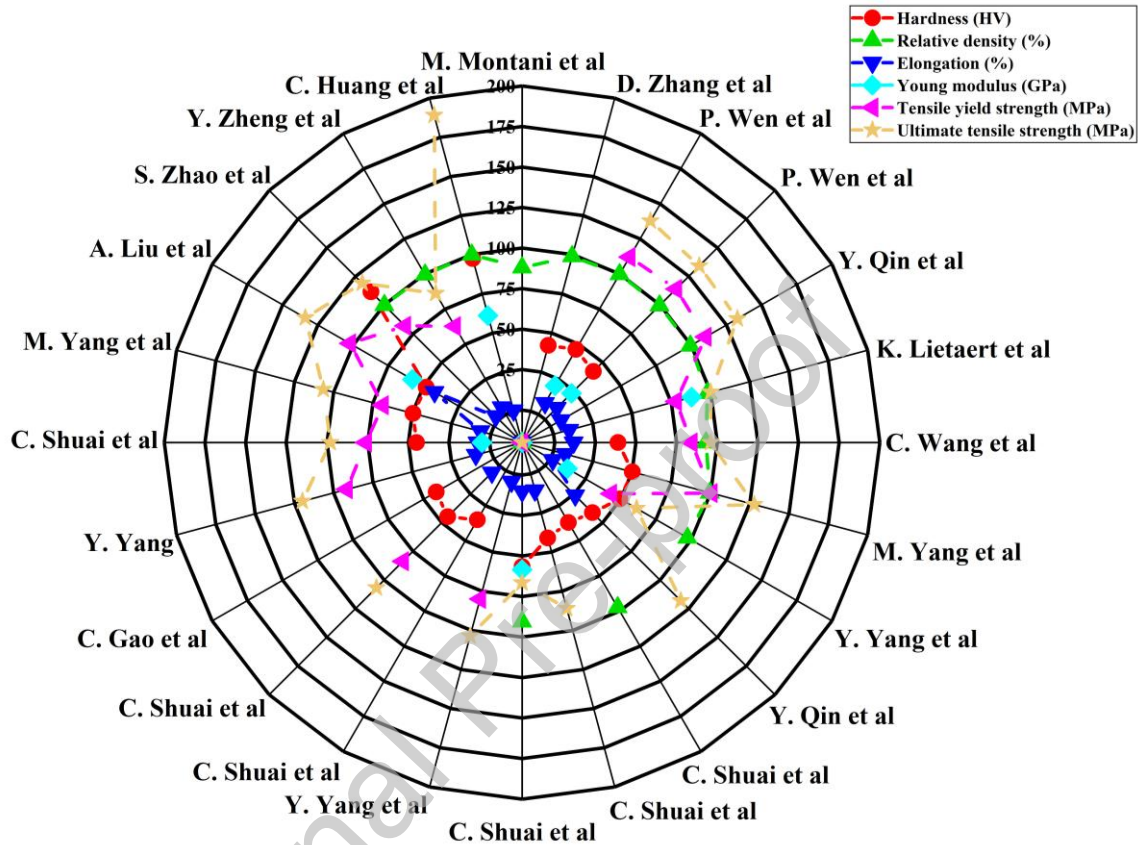


Fig. 11. Mechanical characteristics of Zn and its alloy by the L-PBF process [20,56,60,63,96,100,103,106,123-138].

(1) Tensile property

The L-PBF process features a limited molten pool volume, with rapid movement due to the high energy density of the laser beam. The typical cooling rate during L-PBF exceeds that of normal casting operations ($1-10^2$ K/s), resulting in exceedingly fine to very fine grains [117]. The laser path determines the direction and extent of heat loss, with the sharpest directional thermal gradients (G) occurring towards the cold substrate and the center of the melt pool [106,116,139]. Strategies typically include altering the hatch pattern between levels, such as 67° , 90° , or random orientations, to interrupt ongoing epitaxial growth and mitigate the buildup of residual stresses, which may cause warping [56,106,140]. Following multiple cycles of melting and solidifying during L-PBF, the final grain structure is established. The preferred orientation of the columnar grains gradually shifts towards the BD. The refined grain size and orientation growth along the BD are significant features of the grain structure produced by

standard L-PBF [117]. Grain growth preferentially occurs in the direction of the largest temperature gradient at the solid–liquid interface during solidification. For hcp metals such as Zn and Mg, optimal growth is along $\langle 1010 \rangle$. The scanning speed also influences grain growth and orientation; ongoing preferred growth along $\langle 1010 \rangle$ was observed at a relatively low scanning speed of 300 mm/s, resulting in larger columnar grains [141]. The L-PBF area subjected to multiple heat cycles comprised coarse columnar grains that exhibited optimal growth in the $\langle 0001 \rangle$ direction, similar to pure Zn samples produced at a scanning speed of 300 mm/s. As scanning speed increased, the columnar grains did not have sufficient time to enlarge. Epitaxial growth at a scanning speed of 500 mm/s resulted in finer grains, irregular grain morphologies, and poor grain texture. An increase to 700 mm/s led to a much slower growth rate of columnar grains, yielding smaller grains and more refined polygonal grains with an average size of 5.9 μm [142]. Generally, faster scanning speeds resulted in smaller grain sizes and weaker textures, as illustrated in Figs. 10(a), (b), and (c). The Schmid Factor (SF) governs the relationship between grain orientation (texture) and slip-system activation in zinc by transforming macroscopic stress into resolved shear stress on specific crystallographic planes [33,108]. The Hexagonal Close-Packed (HCP) structure of zinc, characterized by a high c/a ratio (~ 1.856), exhibits limited independent slip systems at room temperature, rendering mechanical properties highly sensitive to the orientation of the crystal lattice relative to the loading direction³³. Zinc may deform mainly via three slip systems: basal, prismatic, and pyramidal planes. The anisotropy of the yield strength is primarily caused by the orientation of the basal plane relative to the tensile axis. In vertical samples, the fast cooling and directed thermal gradients of the L-PBF process cause grains to develop in the building direction in an epitaxial manner. This typically results in a texture where the $\langle 0001 \rangle$ (c -axis) direction is perpendicular to the tensile axis^{63,106}. The c -axis's orientation perpendicular to the loading axis yields a very low resolved shear stress on the basal planes (a low Schmid Factor). Consequently, the simplest slip mode (basal glide) is crystallographically blocked, requiring significantly higher macroscopic stress to initiate plastic flow, resulting in enhanced yield strength³³. By contrast, samples with grains aligned at various angles (0° to 90°) to the tensile axis exhibit a higher average SF for basal glide, yielding at lower stress levels. The schematic diagram is displayed in Fig. 12(a). Qin et al.¹⁰⁶ demonstrated that stress-strain curves for L-PBF samples were generated with varying scanning rates and building directions, as shown in Fig. 12(b). All samples exhibited signs of work hardening upon yielding, with fracture occurring during the tension rise rather than during obvious necking. The stress curves for vertical samples showed peak values significantly higher than those of horizontal samples. Matching tensile characteristics, such as the elongation (δ), ultimate strength (σ_m), and yield strength ($\sigma_{0.2}$), are illustrated in Fig. 12(c). In vertical samples V1, V2, and V3, $\sigma_{0.2}$ and σ_m increased with rising scanning velocity, with the V3 sample exhibiting the highest values of 110.3 ± 2.10 and 132.9 ± 0.70 MPa, respectively. The V1 specimen exhibited the lowest elongation value at $2.60\% \pm 0.50\%$, while the V2 specimen showed the highest elongation value at $8.10\% \pm 0.9\%$. Ductility in horizontal specimens increased with scanning velocity,

while strength remained constant across all scanning speeds. Among the horizontal specimens H1, H2, and H3, specimen H3 displayed the highest ultimate strength (92.3 ± 1.8 MPa) and elongation ($5.0\% \pm 0.9\%$), both of which were lower than the comparable values for specimen V3. Overall, vertical samples exhibited greater strength and ductility than horizontal ones, and a higher scanning speed enhanced both properties¹⁰⁶. Horizontal samples frequently show grains with c-axes aligned at various angles (0° – 90°) to the tensile axis, resulting in an elevated average SF for basal glide that allows the material to yield at reduced stress levels⁶³. Dense L-PBF Zn tensile bars outperformed cast Zn in terms of strength and elongation while exhibiting characteristics inferior to those of extruded Zn⁹⁵. These methods produce distinct microstructures with smaller grain sizes, thereby improving mechanical characteristics [143], as stated by Karel et al. Samples oriented vertically exhibited greater stiffness and strength but demonstrated reduced ductility compared to those oriented horizontally, as shown in Fig. 12(d). Variations in grain orientation may account for the differences in flow stress between vertical and horizontal samples. In vertical samples, the predominant grains were oriented parallel to the tensile direction, leading to a significant decrease in the resolved shear stress of the basal slip system, which then accounted for only a small percentage of the total applied stress. Conversely, horizontal samples showed grains aligned at angles between 0° and 90° with respect to the tensile axis along the $\langle 0001 \rangle$ direction. Consequently, even with the same applied tensile force, certain grains in the horizontal samples experienced greater resolved shear stress on the basal slip system. This indicates that horizontal samples may initiate plastic deformation at lower tensile stress levels than vertical samples⁶³. Similarly, Voshage et al. [144] demonstrated that the tensile specimen, characterized by a lattice structure face-centered cubic with vertical struts (F_2CCZ), fails due to transcrystalline cleavage fracture, with a strut diameter close to $500 \mu\text{m}$, a cell size width of 3 mm , a specimen diameter of 13 mm and a height of 90 mm . When a load of 500 kN is applied during the tensile test at a strain rate of 10^{-3} s^{-1} , the yield strength, ultimate tensile strength (UTS), and fracture elongation are 101.67 MPa , 113.33 MPa , and 10.8% , respectively [144]. Textured (rolled) zinc exhibits enhanced strength when the applied force direction coincides with the majority of its basal planes [145]. The scanning strategy plays a critical role in the thermal gradient, which determines the mechanical and microstructural behavior of the metal or alloy, as shown in Table 4.

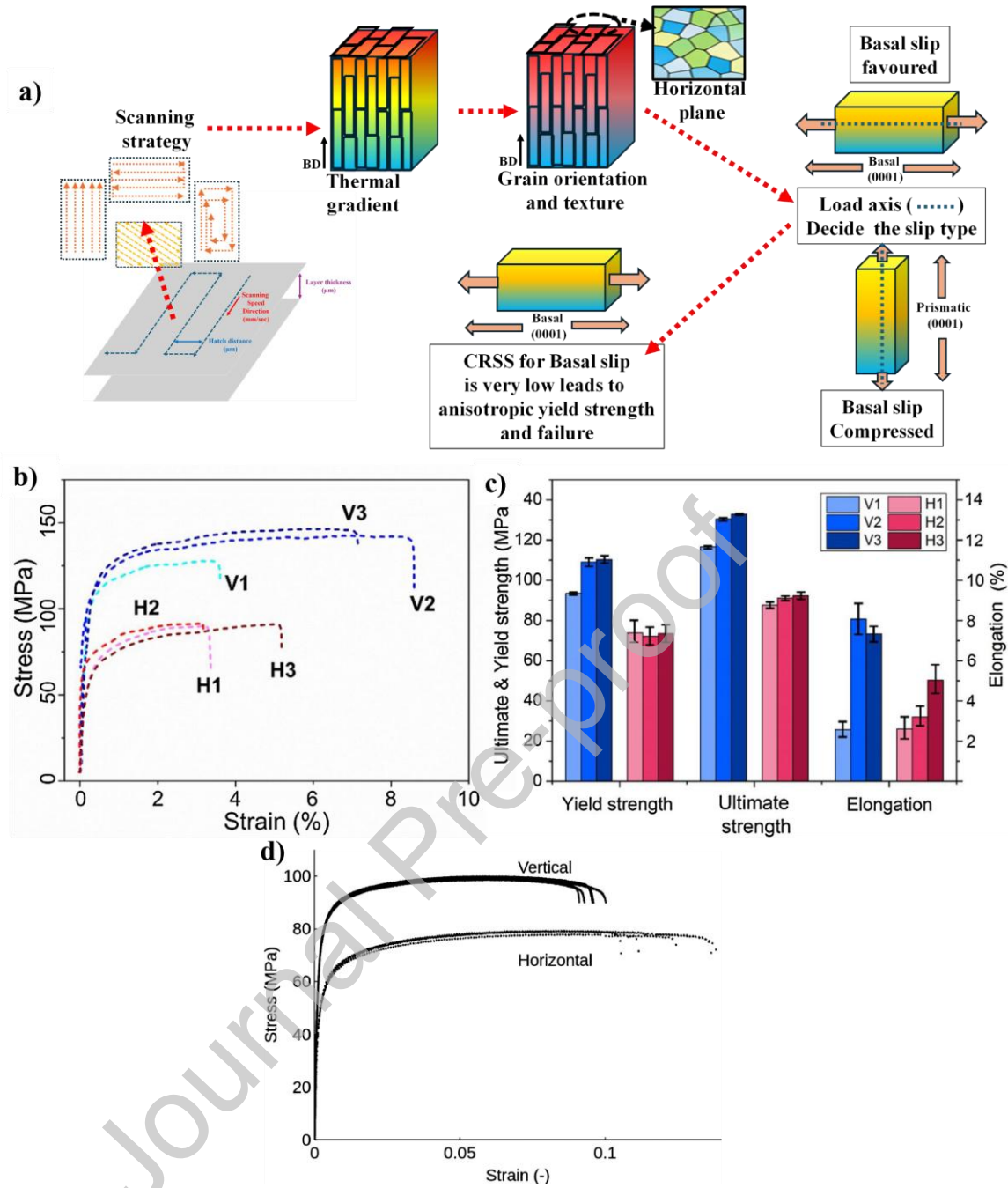


Fig. 12. (a) Scanning strategy effects on anisotropic yielding (b) stress-strain graph, (c) tensile properties, and (d) plot of vertical and horizontal orientation of tensile samples ^{63,106}

(2) Compression property

Karel et al. demonstrated that compression testing of Zn scaffolds produced a uniform stress-strain curve, devoid of abrupt stress declines in the plateau region, as shown in Fig. 13 ⁶³. The density distribution for the L-PBF-fabricated scaffolds and the replicated Zn scaffolds reflects the plateau stress, defined as the average stress within the strain range of 0.2 to 0.3 times, as shown in Fig. 13(a) [146]. The plateau stress of a cellular metal is determined by its density, microstructure, and mesostructure, which refers to the solid-phase architecture at the strut and pore scales [147]. For L-PBF Zn scaffolds, the solidification structure's size and shape are

unlikely to significantly affect strut orientation⁶³. In contrast to AlSi10Mg, an opposite conclusion is suggested: the relationship between microstructure and strut orientation varies by material [148,149]. Although all L-PBF Zn scaffolds exhibit comparable microstructure, the mechanical strength of scaffolds with various unit-cell designs can be examined to determine the impact of the mesostructure. Conversely, LPBF Zn scaffolds with comparable mesostructural characteristics and unit-cell dimensions may be fabricated to investigate the role of microstructure [146]. The nodal connection seen in diamond and dodecahedron unit cells is consistent with this. These structures may be examined alongside solid zinc to determine how strut microstructure affects mechanical strength, as indicated in Fig. 13(b). Kagome and face-centred cubic (FCC) unit cell scaffolds exhibit significantly higher plateau stress compared to other unit cells at equivalent densities, highlighting the structural efficiency of these mesostructures. The enhanced performance can be attributed to greater interconnectedness and the strategic placement of struts at slight angles to the loading direction, leading to a deformation process dominated by stretching. In the 20%-25% density range, the diamond and dodecahedron scaffolds exhibited marginally elevated plateau stress compared to replicated cellular zinc. The role of grain boundary strengthening is crucial in influencing this behaviour, as outlined by the Hall-Petch relationship, where σ_0 and k represent constants specific to the material and d denotes the grain size.

$$\sigma_y = \sigma_0 + \frac{k}{\sqrt{d}} \quad (2)$$

where σ_0 is the original yield stress that is equivalent to 32.4 MPa, and k is 0.22 for Zn [150]. While the exact difference in grain size between the replicated material and L-PBF remains ambiguous, it is estimated to be approximately one order of magnitude. Given that the particle size exceeds the pore size, it is anticipated that the replicated material will exhibit minimal grain boundary strengthening [146]. A flow stress of approximately 66 MPa can be estimated for Zn fabricated using the L-PBF process, considering the base flow stress ($\sigma_0 = 32$ MPa) and the fact that the dimensions of pores and struts in the replicated cellular material exceed the scale at which size-dependent plasticity effects typically occur (i.e., below 10 μm), as shown in Table 3 [63, 151, 152]. Consequently, assuming the mesostructure remains consistent, the strength of the L-PBF-fabricated scaffolds is expected to be approximately double that of the replicated versions. However, this variation exceeds the recorded 20% strength enhancement in L-PBF dodecahedron-based scaffolds compared to replicated Zn at 20% relative density, as well as the 40% increase in diamond-based L-PBF scaffolds. The results indicate that the replicated scaffolds maintain a commendable load-bearing capacity, despite their standard geometry and reduced interconnectivity. Table 3 shows that the scaffolds under examination have lower strength values than those produced from bioinert materials often utilized in clinical practice. Variations in the bulk material's yield strength may account for differences in strength across scaffolds with identical unit cells and densities. The yield strengths of pure titanium, pure tantalum, and Ti6Al4V, treated using L-PBF, are approximately 350 MPa, 650 MPa, and 850 MPa, respectively [153]. Despite dense L-PBF Zn exhibiting a yield strength of approximately 66 MPa, it can be inferred that, given an identical mesostructure and relative

density, the plateau stress of L-PBF Zn scaffolds is expected to be about 19% of that of Ti scaffolds, 10% of Ta scaffolds, and 8% of Ti6Al4V scaffolds. This computation is underpinned by the direct relationship between strut yield strength and scaffold plateau stress. The validation of this prediction is achieved by comparing Zn and Ti scaffolds with a dodecahedron unit cell and 30% density. In evaluating the various materials and unit cell types, it is evident that Zn scaffolds exhibit higher base-metal flow stress. Direct comparison of plateau stresses presents challenges, as Ti6Al4V scaffolds exhibit less consistency in compressive stress-strain behavior compared to those made from pure Ti, Ta, or Zn. The strength measurements from biodegradable iron scaffolds may serve as a basis for further comparisons [154].

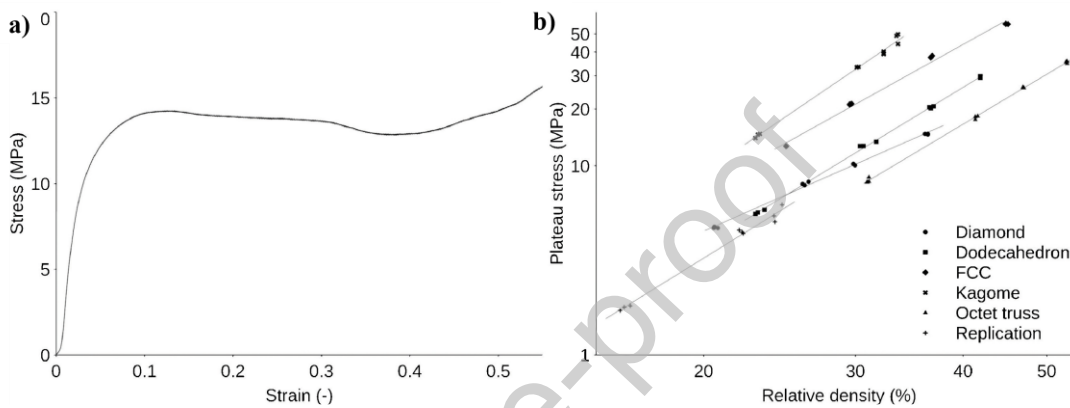


Fig. 13. (a) A Kagome unit cell scaffold with a relative density of 23% exhibits a smooth stress-strain response under compressive loading ^{63,146}, (b) A log-log plot illustrates the correlation between plateau stress and density for L-PBF Zn across various unit cell designs, alongside replicated Zn for comparative analysis ^{146,150}.

Table 3. Analysing the recorded plateau stresses for various porous structures that share identical relative density and unit cell configurations.

S. No.	Unit cell	Material	Relative density	Plateau stress	Ref. No.
1.	Diamond	Ti6Al4V	30	50	[151]
2.	Diamond	Zn	30	10	
3.	Dodecahedron	Zn	20	6	[63]
4.	Dodecahedron	Zn	30	30	
5.	Dodecahedron	Ta	20	25	[152]
6.	Dodecahedron	Ti	30	60	[151]

2.2 Defects in pure zinc and its alloys

Surface imperfections in the L-PBF process are related to the complex thermodynamic features of the chaotic molten pool, governed by process parameters [155,156]. The high evaporation rate of Zn results in significant, irregular vapour flow within the melt pool [96,157], increasing the uneven flow of molten metal and leading to porosity, balling, fractures, and alloying element depletion. Un-melted powder particles and porosities are common in L-PBF components, irrespective of scanning speed or method. Surface sintering necks indicate inadequate laser energy to melt Zn powder. As L-PBF power is increased, the melt trace

stabilises and becomes continuous. With increased laser power, track width steadily increases and approaches the merger point with adjacent tracks ¹²⁴. The elevated temperature from strong laser intensity significantly reduces the melt's surface tension, reinforcing the metallurgical connection between adjacent tracks. Optimisation of the L-PBF method is crucial for achieving overlapping, efficient metallurgical bonds between tracks, resulting in smooth, flat, stable tracks and superior surfaces. Moreover, significant Zn evaporation may increase the rebound pressure within the molten pool, causing the melt to fragment into minute droplets and producing a balling effect. The interplay between high mass transfer and melt instability leads to the emergence of micro-humping events on the solidified surface. A decrease in solute in the molten pool results in discontinuous scanning traces [158]. The density of the part depends on the powder melting behavior, including the volume of flowing liquid metal and the balance between evaporation and keyhole formation. Insufficient laser energy leads to non-fusion of passes and layers. Continuous decline in E_v worsens density, with non-fusing pores often being non-round and located in the corners of passes and layers. Laser energy also influences Zn evaporation by heating the molten pool, thereby accelerating evaporation. The keyhole becomes less stable as evaporation increases the Zn vapor concentration, occasionally causing the keyhole to move during L-PBF. In high- E_v regions, the keyhole collapses, trapping Zn vapor in the molten pool and forming pores. Random round pores result from gas entrapment, and at elevated E_v , removal of evaporation fumes may become challenging. With each layer, surface deterioration occurs, leading to decreased powder quality due to increased surface roughness and resulting in the bursting of solidified materials after several layers. With sufficient laser energy, complete melting of all powders occurs, and the keyhole remains stable. Due to keyhole oscillation and repeated melting and solidification, gas entrapment holes cannot be entirely eliminated. Gases such as air, metal vapor, or powder may be produced. If H_s and D_s change, the optimal E_v value may change as well ⁵⁶. The shape and size of the powder also affect porosity and defect type at the same volume energy density. Finer powder exhibits greater sensitivity to influence levels. Spierings et al. demonstrated that the use of finer powder facilitates the attainment of fully dense stainless-steel pieces, as the required energy density is lower compared to larger powder sizes [159]. Demir et al. found that fine Zn powder (9 μm) exhibits higher porosity at the same volume energy density than coarse (15 μm) or large powders. The porosity and agglomeration of LPBF-processed zinc are influenced by laser energy density and powder particle size. These factors affect melt pool stability, evaporation, and densification. Insufficient melting (below 40 $\text{J}\cdot\text{mm}^{-3}$) leads to lack-of-fusion faults, resulting in irregular, non-circular holes between scan tracks and layers. This phenomenon occurs with both types of powder, although under-melting is more pronounced with coarse powder. As laser energy density increases, melt stability improves, thereby reducing porosity. The stability window is significantly affected by powder particle size. The processing regime for coarse particles (5-100 μm) is broad ($\sim 40\text{-}115 \text{ J}\cdot\text{mm}^{-3}$), resulting in relative densities exceeding 99%. Smaller, circular holes indicate that pore growth primarily occurs post-solidification, rather than during partial fusion. Fine particles are more sensitive to variations

in laser energy density. Due to the larger surface area, Zn vaporizes more rapidly with increased energy. At high layer numbers, the melt pool becomes unstable, leading to particle ejection and material explosion. High fluence ($> \sim 115 \text{ J} \cdot \text{mm}^{-3}$) may induce balling-like instabilities and catastrophic material rupture, particularly in fine powders, due to elevated recoil pressure and heat from evaporation. These findings demonstrate that energy density and powder particle size significantly influence LPBF zinc porosity and balling, impacting vaporization kinetics, melt pool stability, and the parameters of the defect-free processing window. As particle size decreases at a constant volume energy density, the vaporization sensitivity of Zn powder increases⁵⁹. The types of defects and their effects on mechanical properties are presented in Table 4 [16, 18, 56,63,106, 119,124, 160-188].

Table 4. Process - Microstructure - Property Relationships of Zinc.

S.No	Scanning strategy	Laser parameter	Microstructure (Grain morphology)	Mechanical properties	Defects	Explanation	Ref. No.
1.	90° rotation between layers	P : 40-120 W, V : 200-1000 mm/s, Hs : 70 μm , Ds : 30 μm .	Shape: Fine columnar grains	YS: 114 MPa, UTS: 134 MPa, Elongation: 10.1%, Hardness: 42 HV.	Irregular lack-of-fusion cavities and circular gas entrapment pores	Rapid cooling and directed heat gradients promote fine columnar development. High scanning speed limits growth time, leading to refined, randomly oriented grains.	56
2.	90° rotation between layers	P : 80 W, V : 300-700 mm/s, Hs : 70 μm , Ds : 30 μm .	Shape: Transition from coarse columnar to refined polygon grains at high speed.	YS: 110.3 MPa, UTS: 132.9 MPa, Elong: 8.1%.	Random tiny pores (1-10 μm) from keyhole instability.	Increasing scanning speed reduces the temperature gradient, increasing thermodynamic undercooling	106
3.	90° alternating rotation	P : 80-100 W, V : 400-800 mm/s, Hs : 55 μm , Ds : 45 μm .	Shape: Refined equiaxed grains at high speeds.	YS: 100.7 MPa, UTS: 127.8 MPa, Elong: 7.6%, Hardness: 50.2 HV.	Spherical particles (balling), micro-humping and keyhole pores.	Mechanical alloying creates nano-grains; ultra-fast LPBF cycles inhibit growth, preserving heterogeneity	124
4.	90° raster angle	P : 85 W, V : 180 mm/s, Hs : 50 μm .	Shape: Micron-grain cores with nano-grain Shell, Core/shell heterogeneous. Size: Core $\sim 6.4 \mu\text{m}$, Shell $< 50 \text{ nm}$, Columnar grains at the bottom and equiaxed in the middle and top.	CYS: 137.9 MPa; Plasticity: 58% improvement.	Extra evaporation of nano-grains during L-PBF.	Heat accumulation in the middle region triggers dynamic recrystallization.	18
5.	Zig-Zag 67° rotation	P : 40-60 W, V : 150 mm/s, Hs : 70 μm , Ds : 30 μm .	Shape: Zonal equiaxed and lamella, Size: 12.7-15.5 μm .	YS: 82.02 MPa, UTS: 116.20 MPa, Elong: 37.16% (at 60 W)	Tiny holes and conical keyholes.		141
6.	45° and 135° rotation	P : 80 W, V : 900 mm/s, Hs : 55 μm , Ds : 30 μm .	Shape: Bimodal: Alternating fine equiaxed and coarse columnar grains. Size: 9.63 μm (hori), 16.78 μm	UTS: 123.6 MPa (Horiz). Elong: 14.2% (Vert)	Unfused cracks, gas pores, and keyholes.	Alternating tracks create fine grains at track intersections and coarse grains within tracks.	[33]

S.No	Scanning strategy	Laser parameter	Microstructure (Grain morphology)	Mechanical properties	Defects	Explanation	Ref. No.
			(vert).				
7.	Random rotation of hatch patterns. Lattice structures	P : 50-250 W, V : 400-1500 mm/s, H s: based on 40% overlap.	Weakly textured grains growing epitaxially.	E : 110 GPa (Vert), 81 GPa (Horiz); UTS: 100 MPa (Vert)	Median pore size of 3.7 μ m, dross and particle attachment.	Preferred orientation of the <0001> direction in the building plane dictates stiffness.	[63]
8.	Auxetic structures	P : 65-85 W, V : 840-974 mm/s, T_0 : 353-513 K (Pre-heat)	Shape: Elongated columnar grains, Size: 28.7-83.7 μ m.	YS: 0.06-3.24 MPa (depending on auxetic design), UTS: 0.09-4.22 MPa	Surface roughness; microscale defects at feature boundaries.	Directional solidification is driven by heat flux toward the substrate, aligning grains with the building direction.	[187]
9.	TPMS (I-WP, Primitive, Gyroid)	P : 125 W, V : 700 mm/s, H s: 80 μ m, D s: 30 μ m.	High relative density (98.2-99.1%) equiaxed grains.	Compressive yield strength I-WP: 36.1 $\pm$ 1.2 Mpa, Primitive: 33.5 $\pm$ 1.4 Mpa, Gyroid: 21.6 $\pm$ 1.8 MPa Diamond: 20.8 $\pm$ 1.9 MPa, Fischer Koch: 18.6 $\pm$ 1.8 MPa.	Micro-pores from evaporation; non-uniform wall expansion.	Dynamic recrystallization occurs during printing due to heat accumulation.	[119]
10.	Voronoi-based (VBCC, VROC, VBFCC)	P : 80 W, V : 600 mm/s, H s: 55 μ m, D s: 30 μ m.	Struts containing internal micropores.	Yield strength: 3.35-17.59 MPa, Young modulus: 0.31-1.18 GPa	Lack of fusion and vaporization-induced surface micropores	Ordered lattice seeds transform disordered patterns into structures with uniform stress. Face-centered concave nodes allow for tensile expansion and compressive contraction. High cooling rates produce grains significantly finer than cast counterparts.	[103]
11.	Reinforced NPR (Auxetic Type B/C)	P : 60 W, D s: 30 μ m.	Face-centred concave nodes, Shape: Concave strut network.	Yield Strength: 2.3-15.3 MPa (B3: 15.3 MPa), E-Modulus: 85-1327.17 MPa.	Un-melted powder particles, blocked horizontal pores.		[188]
12.	Diamond Lattice (Bone Scaffold)	P : 80 W, H s: 70 μ m, D s: 30 μ m.	Shape: Diamond structure with refined grains, Size: 5.9 μ m.	CYS: 14.9 $\pm$ 1.8 MPa, UCS: 22.9-49.2 MPa.	Evaporation-induced voids, splashing and rough surfaces.		[16]

2.2.1 *Balling effect*

The formation of discrete metal spheres characterizes the balling process observed after laser irradiation. This phenomenon results from the influences of surface tension, gravity, and the surrounding environment on molten metal^{160,161}. The balling effect represents a notable surface imperfection in L-PBF of zinc alloys, as shown in Fig. 14(a). It is also a crucial factor contributing to additional defects^{20,162,163}. This phenomenon can be explained by the Plateau-Rayleigh instability theory, which posits that deposited metallic tracks or pathways disintegrate into hemicylindrical or globular configurations under specific conditions¹⁶⁴. Various process factors, including scanning speed, laser intensity, viscosity, and surface tension of the melted metal, influence the probability of balling¹⁶⁵. The dimensions of balls typically range from approximately 500 μm for ellipsoidal configurations to about 10 μm for spherical forms. Inadequate laser energy frequently results in the formation of larger ellipsoids. The solid Zn alloy does not achieve complete wetting by the liquid melt at these conditions, leading to an uneven distribution of the melted Zn alloy on the baseplate or any pre-existing layer. The interaction between the liquid and solid phases is constrained, as evidenced by a contact angle often exceeding 90° between the deposited layer and the solidified surface. According to the laws of Gibbs free energy, metals tend to form spherical shapes, thereby reducing surface energy^{56,166}. Larger ellipsoidal balling forms may occur due to insufficient wettability between the liquid and solid phases. The primary particles of Zn alloy powder are typically significantly smaller than the larger agglomerated formations observed. The surface quality and dimensional accuracy of Zn alloy specimens are affected by increased melt accumulation height, which may hinder homogeneous powder dispersion. Continuous application of high laser energy frequently results in a secondary effect characterized by the formation of spherical droplets on the molten material's surface, as shown in Fig. 14(b). High energy input often leads to overheating and melt pool vaporization^{167,168}. Elevated temperature conditions produce substantial rebound pressure beneath the laser beam, resulting in an immediate increase in the melt pool's internal temperature. A significant amount of metal vapor is ejected in a splattering pattern as recoil pressure enhances surface vaporization in the melt pool. As the process progresses, non-fused metal particles are conveyed by hot gas flow, resulting in dispersed powder adhering to the metal surfaces, as shown in Fig. 14(c). Additionally, the weld or melt pool absorbs extra surface energy from the high laser energy input, greatly improving flow characteristics. Due to surface-driven melt-pool circulation caused by uneven surface energy distribution, a portion of the molten material is expelled from the melt pool, forming splattered droplets^{169,170}. When these droplets contact the solidification layer, they quickly cool, forming miniature metal spheres. As a result of their diminutive size, the development of tiny droplets during the L-PBF process is often inevitable.

2.2.2 *Pore formation*

Gas pores, spherical pores, keyholes, and incomplete-fusion holes represent typical flaws in Zn alloys produced by L-PBF, as shown in Figs. 14(d) and 14(e). Incomplete fusion pores arise from insufficient metallurgical bonding, resulting from inadequate energy input. Insufficient

energy input leads to inadequate melting of the Zn alloy powder or restricted amalgamation of the molten metal with the hardened substrate ^{140,171}. These voids are several hundred micrometres wide, exhibit uneven geometries, and are most commonly present at the contact points between adjacent tracks of molten material ¹⁷². Insufficient laser energy input produces melt pools that are too narrow, impairing the overlap between adjacent melt tracks. Un-melted powder can remain at the boundaries between tracks when this overlap is insufficient, potentially leading to pore formation. As the build continues, these initial defects can propagate and develop into larger, multi-layer pore imperfections ^{173,174}. Overexposure to laser radiation can cause keyhole defects in the Zn and its alloy substrate ¹⁷⁵. When irradiated with a concentrated laser, powdered zinc alloy can experience localized temperatures above its boiling point. This vaporization produces gas bubbles and a downward reactive force on the melt pool, initiating keyhole formation ¹⁷⁶. Variations in temperature on the molten surface, caused by changes in laser trajectory and energy absorption within the keyhole, generate forces including surface tension gradients, recoil pressure, and Marangoni convection. These events substantially affect keyhole behavior, frequently leading to instabilities and, at times, collapse ^{177,178}. Keyhole collapse in Zn structures can result in the entrapment of voids by the advancing solidification front, making them part of the final build. Additionally, when the keyhole collapses, it can reduce the volume of the surrounding cavity, producing vapor-encapsulated pockets. As the keyhole reopens, some of these bubbles may be reabsorbed; however, most remain trapped as the solidification front advances at the keyhole base, ultimately resulting in the formation of numerous spherical pores within the Zn layer ¹³³. Efforts to address keyhole defects usually involve reducing laser energy input, typically by increasing scanning speed or reducing beam power ^{123,139,179,180}. Reducing ambient pressure may mitigate keyhole formation. Consequently, this reduction may impede the ability of Zn alloy powder to effectively infiltrate the molten pool during the L-PBF process, thus compromising the overall quality of the produced component ¹⁸¹. No proven method exists for completely eliminating keyholes by altering laser beam processing input variables or parameters; however, studies indicate that the use of nanoparticles may help reduce keyhole formation. As a result, the laser input variable and processing parameters for zinc alloys are usually limited to a narrow working window to prevent keyhole formation. This limitation renders the L-PBF process highly vulnerable to changes in energy input and prone to flaws such as pores and keyholes ^{103,180}. Furthermore, the inner subsurface of these pores exhibits significant roughness, suggesting hydrodynamic turbulent flow within the molten pool. Microscopic pores are inherent to the L-PBF process due to high laser energy or beam density. These cannot be entirely removed without supplementary interventions, such as hot isostatic pressing or post-processing methods.

2.2.3 Residual stress and cracking

Residual stress poses a significant challenge in L-PBF, primarily due to the steep temperature gradients and rapid cooling rates characteristic of this technology ^{63,182}. The accumulation of these stresses in the alloy may cause structural flaws, including cracks, distortions, warping, and delamination, which significantly affect construction quality and adherence to strict

dimensional standards. In L-PBF, internal stresses can be categorized into two primary types, each stemming from a distinct mechanism of formation ^{183,184}. Residual stress develops from the temperature differential in the newly formed layer and the thermal contraction occurring as the molten upper layer cools. The focused energy of the laser induces thermal expansion at the surface of the freshly deposited material. This expansion is constrained by the cooler, previously deposited underlying layer, resulting in compressive tension at the surface. Exceeding the yield strength of the zinc alloy due to thermal expansion may lead to plastic deformation in the solid layer. Upon cooling, the areas subjected to plastic deformation contract and revert, transforming the initial compressive stress into residual tensile tension. The second mechanism indicates that the liquid layer retains a significantly higher temperature than the cemented substrate beneath it. As the upper layer cools and solidifies, it experiences substantial contraction. The reduction in dimensions is constrained by the underlying solid layer, resulting in tensile stress in the top region and compressive stress in the lower layer. Variations in stress distribution are critical in the development of faults during LPBF processing of zinc alloys. Principal cracking processes observed in this process include solidification-induced cracking and liquation microcracks ¹⁸⁵. Hot tearing, also referred to as solidification microcracks, typically occurs during the final stages of solidification. The formation of these cracks is exacerbated in L-PBF by steep temperature gradients that inhibit adequate liquid movement between dendrites ¹⁸⁶. As the material cools and the liquid metal becomes more viscous, microcracks may form in the fusion zone or melt pool during solidification. The restriction of liquid metal flow is intensified by the development of rigid dendritic formations. Microcracks formed during the liquid phase are evident during the solidification of Zn alloy materials. The rapid cooling rates associated with L-PBF may induce segregation of alloying elements, potentially resulting in localized regions with lower melting points. Continued exposure to elevated temperatures may cause re-melting of these regions, particularly near the boundaries of previously solidified areas. If the remelted regions fail to withstand thermal stresses, microcracks attributable to the liquid are likely to develop around the melt pool. As liquation microcracks form, they become sites for further microcrack propagation, progressively enlarging as the Zn alloy continues to deposit. Microcracks often occur in zinc alloys with numerous alloying elements or when substantial differences exist in the coefficients of thermal expansion between the nanoparticles and the Zn matrix, which may result in unstable melt tracks, as illustrated in Fig. 14(f) ⁹⁴.

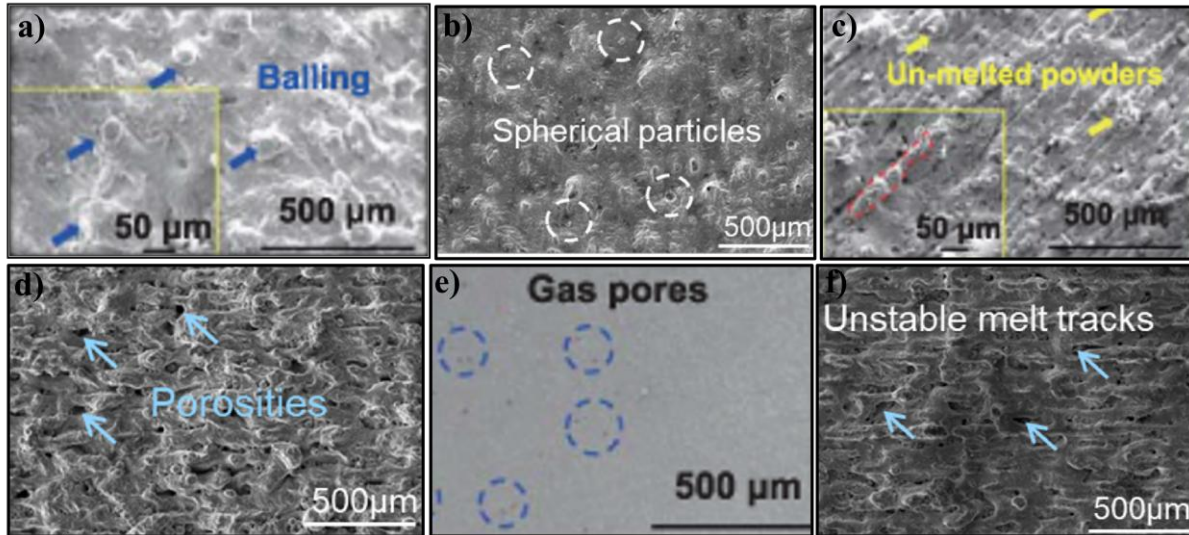


Fig. 14.Types of defects in pure Zinc (a) Balling effect ¹⁰³, (b) Spherical particles ¹⁰⁸, (c) Unmelted powders ¹⁰³, (d) Porosities ¹⁰⁸, (e) Gas pores ¹⁰³, and (f) Unstable melt tracks ¹⁰⁸

2.3 Zinc and its alloys' suitability for orthopaedic implants

2.3.1 Mechanical characteristics

The physicochemical properties of materials, including mechanical properties and corrosion rate, play a crucial role in assessing compliance with standards for internal fixation implants. Load-bearing implants are advised to possess an UTS exceeding 0.3 GPa and an elongation (ϵ_f) ranging from 0.15 to 0.18 times ¹⁴². However, pure zinc exhibits a Young's modulus of less than 0.2 GPa ⁸³, which is inadequate for optimal orthopedic internal fixation implants. Recent research has shown that the introduction of alloying elements (Mg, Li, Ag, Cu, etc.) can significantly increase the mechanical strength and flexibility of alloys despite the low strength of pure zinc ¹⁸⁹. The kinetics of the alloying system, influenced by the type, amount, and number of alloying components, differentially contribute to the strengthening processes of zinc alloys. Alloying with Mg results in the greatest increase in UTS, while alloying with Mn and Cu yields Zn-based alloys with the highest ϵ_f values. Under the same processing conditions, the addition of a new component to the Zn binary alloy improves UTS while decreasing ϵ_f . Conventional methods such as extrusion, rolling, and drawing enhance mechanical properties, resulting in higher UTS and ϵ_f than as-cast alternatives. Hydrostatic extrusion is recognized as the most effective method for severe plastic deformation (SPD), particularly concerning UTS and elongation at fracture. The remarkable grain refinement associated with SPD is well established to enhance mechanical properties and corrosion resistance ¹⁹⁰. Although pure zinc still lacks adequate strength, its high elongation (up to 38%–40%) allows for further thermal deformation processing, enhancing the mechanical characteristics of Zn alloys through AM, powder metallurgy, thermal deformation, alloying, and other techniques ^{100,191}. Furthermore, considering the position and function of the implant, a broader perspective is essential for evaluating both elongation and tensile strength parameters for load-bearing implants used for internal anchoring. The mechanical criteria for materials used in orthopaedic implants are determined by anticipated operational conditions and specific applications. The five essential mechanical parameters are Young's modulus, yield strength, UTS, fracture toughness, and elongation at break. Bone replacements must provide adequate biomechanical strength, particularly in the tibia and femur, as these areas endure significant weight-bearing stresses. The biomechanical strength of an ideal bone substitute must correspond with that of the missing bone. The compressive strength of cancellous bone ranges from 2 to 38 MPa, while human cortical bone exhibits a compressive strength between 130 and 290 MPa and a tensile strength of 90–190 MPa. The durability of Zn-based implants must meet the mechanical demands of both cancellous and cortical bone. Previous research indicates that the optimal orthopaedic implant should demonstrate an elongation of 15%–18% and an UTS exceeding 300 MPa ^{192,193}, sufficient to meet the mechanical requirements of implants in the femur and tibia, two load-bearing zones. In cyclic load-bearing situations, fatigue strength is a critical mechanical criterion for orthopaedic implants in hip, knee, and ankle replacements. The loading stress level following total hip replacements may exceed the patient's body weight significantly. Complete hip replacements are expected to last 20 years while sustaining the body's structural integrity

despite experiencing loads of up to 300 MPa approximately 10^7 cycles¹⁹⁴. Fatigue, also referred to as fretting corrosion fatigue, denotes a state of exhaustion that may occur alongside stress and degradation in the body. Given that fatigue fracture is a primary cause of early failure in biomedical implants, materials with exceptional fatigue strength must be employed for orthopaedic implants. The mechanical property specifications for Zn-based implants must vary depending on the specific insertion site. In the design of Zn and its orthopaedic implants, the relationship between mechanical strength and elastic modulus is critical. A scaffold must achieve a balance between these two traits to be effective in biomedical applications; it must provide sufficient support to regenerating bone while maintaining a stiffness similar to that of natural tissue to prevent stress shielding. Compared to the different types of bones in the human body, the LPBF of Zn and its alloy exhibits a favorable balance between strength and elastic modulus, as illustrated by the Ashby diagram in Fig. 15.

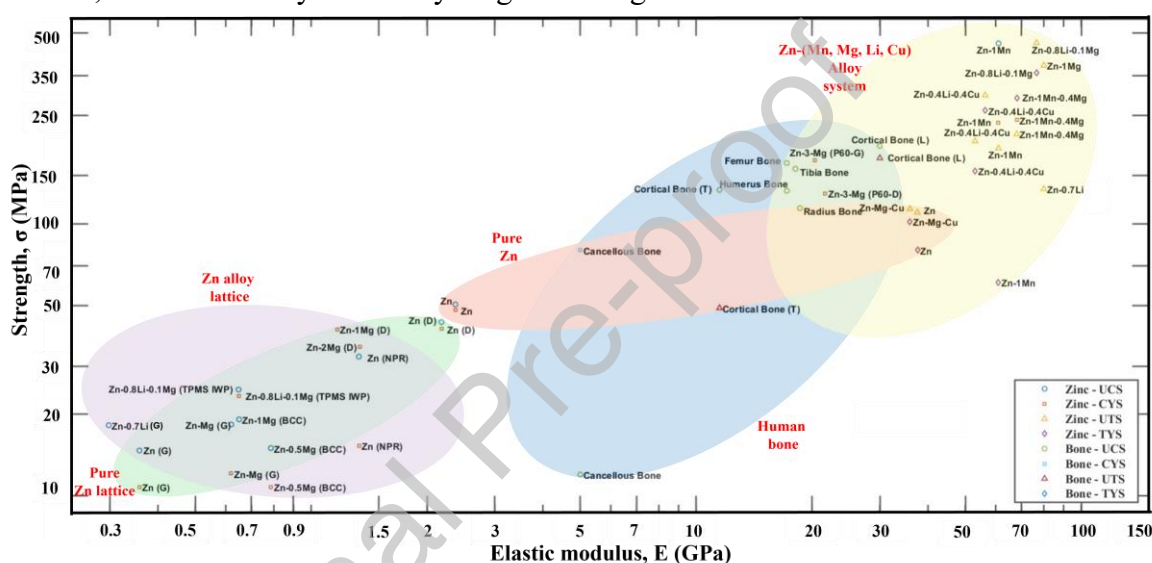


Fig. 15. Strength (MPa) versus Elastic modulus (GPa) of the bone and Zn and its alloys
17,125,130,134,137,138,195-201

2.3.2 Biocompatibility studies

The biocompatibility of materials used in internal fixation implants is essential to in vivo biological activity. Zn is a trace element required by humans in amounts of approximately 2–11 mg daily²⁰². Zinc is vital for several physiological processes, including nucleic acid metabolism, signal transduction, gene expression, apoptotic regulation, and endocrine control. The cellular effects of Zn-based implants are primarily determined by the release of Zn^{2+} from the implants. Studies indicate that zinc supports bone health in various ways. Zn inhibits osteoclast activity while promoting the differentiation, proliferation, and mineralization of osteoblasts²⁰³. Zn^{2+} is necessary for cell division, and its deficiency may cause osteoblastic cells to undergo apoptosis via a mitochondria-mediated mechanism²⁰⁴. Mitochondria are crucial for regulating cell death, and mitochondrial dysfunction is considered an initial event in apoptosis²⁰⁵. According to Guo et al. [204], increased mitochondrial Bax expression

suggests that zinc (Zn^{2+}) deficiency facilitates the transfer of Bax from the cytosol to the mitochondrial membrane. This transfer leads to a significant drop in mitochondrial membrane potential. The movement of Bax from the cytoplasm to mitochondria is critical for cell death, as Bax may reduce mitochondrial membrane potential, increasing permeability and releasing AIF and cytochrome ²⁰⁶. The response of BMSCs to Zn^{2+} is concentration-dependent, resulting in diverse cellular effects. Recent studies have identified "double-edged sword" effects of Zn^{2+} microenvironments on rBMSCs (rat) through various mechanisms. Optimal Zn^{2+} concentrations ($2\text{--}5\ \mu\text{g mL}^{-1}$) enhance rBMSC adhesion and proliferation by up-regulating integrin $\alpha 1$ (ITGA1) and $\beta 1$ (ITGB1, CD29) expression. Importantly, these ideal Zn^{2+} microenvironments modestly increase cellular Zn^{2+} levels by promoting ZIP1 and ZnT1 expression. Activation of the MAPK/ERK signaling pathway and increases in bone formation markers (ALP, Col-1, and OPN) also differentiate rBMSCs into osteogenic cells. Conversely, high Zn^{2+} concentrations ($15\ \mu\text{g mL}^{-1}$) hinder cell adhesion and proliferation by decreasing integrin expression. Elevated ZIP1 expression results in excessive Zn^{2+} influx into rBMSCs, leading to inhibition of osteogenic differentiation and cell death via ROS generation. This mechanism underlies the dual effects of Zn^{2+} microenvironments on BMSCs. Zinc-doped biomaterials may promote bone growth and offer safety for living organisms following this discovery ²⁰⁷. Additionally, Zn facilitates osteoblast protein synthesis by initiating tRNA production and enhancing the function of genes involved in these processes. Increasing the amount of DNA in cells aids osteoblast growth and mineral formation, while altering calcium ion influx can slow osteoclast activity ²⁰⁸. Thus, Zn contributes to new bone formation by enhancing osteoblast mineralization and decreasing osteoclast activity, with a lower risk of disrupting bone chemistry compared to other essential trace elements. Its levels in the body correlate closely with enzymes, proteins, and factors related to bone function. Apart from that, it is very important for protein storage and for controlling key enzymes such as alkaline phosphatase, collagenase, and aminoacyl-tRNA synthetase ²⁰⁹. Moreover, zinc deficiency can lead to osteopenia, impaired bone growth, and other bone-related issues. Zn plays a critical role in the bone environment and significantly affects bone chemistry. However, due to competition with copper for binding to metallothionein, excessive zinc can result in copper deficiency (Cu) ⁹².

2.3.3 Corrosion studies

Zn is extensively utilized for corrosion prevention in maritime and industrial applications ²¹⁰. In these sectors, metallic goods are coated with Zn-based mixtures or compounds, serving as sacrificial coatings to inhibit corrosion of structural components. The primary parameters assessed in in vitro (outside the biological environment) and in vivo (within animals or humans) testing are metal implant accessibility, biocompatibility, and biodegradability. Implants must degrade in vivo for biomedical use as temporary replacements for medical reasons ^{211,212}. Significant in vitro biological corrosion research includes immersion and electrochemical polarization testing (e.g., potentiodynamic polarization experiment, PPE). This approach employs a three-electrode configuration consisting of an opposing electrode (e.g., platinum), a

reference electrode (e.g., a standard electrode), and a working electrode (e.g., zinc). The electrolyte mimics physiological conditions, such as phosphate-buffered saline (PBS), simulated body fluid (SBF), and equilibrium salt solutions like Ringer's (RBSS) and Hank's ²¹³. In electrochemical tests, a voltage is applied to the working electrode, and the resulting current is measured. The set scanning rate evaluates the corrosion current density (i_{corr}) and corrosion potential (E_{corr}). The corrosion rate can be defined through the given Eq. (3).

$$CR = 3.27 \times 10^3 \frac{i_{corr}EW}{\rho} \quad (3)$$

The density of the metal is ρ , EW is the equivalent weight, and the i_{corr} is the current density. Another method is electrochemical impedance spectroscopy (EIS) ²¹⁴. Traditionally, electrochemical tests use direct current polarisation, whereas electrochemical impedance spectroscopy (EIS) utilises alternating current (AC) polarisation. It can evaluate the i_{corr} by polarisation resistance R_p and the Stern-Gerry relationship:

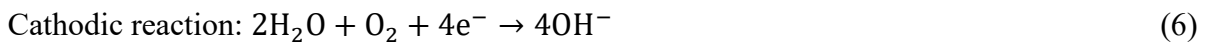
$$i_{corr} = \frac{\beta_a \beta_c}{2.303 R_p (\beta_a + \beta_c)}, \quad (4)$$

where β_a and β_c are the anodic and cathodic Tafel slopes, the value of the i_{corr} is used to evaluate the CR by putting it in Eq. (4).

Electrochemical testing requires immersion of the specimen in a corrosive environment for a predetermined duration, typically utilizing physiological fluid under constant flow conditions. The presence of Zn in the solution increases the pH due to its corrosive properties ²¹⁵. This facilitates the measurement of the metal's corrosion rate. After the soaking period, the sample is removed from the corrosive medium, and the corrosive materials are eliminated from the workpiece to assess the corrosion-induced weight loss. The corrosion rate, CR (mm/year), is computed using Eq. (5).

$$CR = K \frac{W}{Atp} \quad (5)$$

The surface area is $A(\text{cm}^2)$, the mass loss of the specimen is W (g), K is a constant, the time of exposure is t (h), and the density is ρ ($\text{g}\cdot\text{cm}^{-3}$). The following reaction Eqs. (6), (7), (8), and (9) control the corrosion mechanism of zinc regardless of the corrosive media used in these investigations ^{216,217}.



One significant advantage of zinc is that, unlike magnesium, it does not release hydrogen gas during biodegradation, as demonstrated by comparable processes ^{218,219}. The main byproducts of zinc corrosion include oxides, phosphorus (P), calcium (Ca), chlorine (Cl), phosphates (PO_4), and bicarbonates (HCO_3) ^{88,95,220}. The degradation rates of zinc and its alloys are illustrated in Fig. 16. According to Xia et al. ¹⁶, the short-term breakdown patterns of bulk Zn and pure Zn scaffolds, determined by electrochemical testing, are shown in Fig. 17(a). A

plateau in the anodic region of the polarisation curves for both samples indicates passivation during corrosion. The CR for the pure Zn specimen or scaffold measured at 1.09 ± 0.02 mm/year, with a corrosion potential (E_{corr}) of 1.29 ± 0.01 V, and i_{corr} of 71.39 ± 0.91 $\mu\text{A}/\text{cm}^2$. The scaffold exhibited a notably reduced corrosion potential, increased corrosion current density, and an elevated corrosion rate compared to bulk zinc. Electrochemical impedance spectroscopy (EIS) results indicated that the semicircle diameter of the pure Zn specimen or scaffold was smaller than that of solid or bulk Zn, demonstrating reduced corrosion resistance. Findings revealed that during the initial phases of immersion, the pure Zn specimen displayed increased reactivity in Hank's solution. Similarly, in Fig. 17(b), to evaluate deterioration behavior, immersion experiments were conducted. The pure Zn specimen disintegrated rapidly during the initial immersion phase, leading to an increase in the pH of Hank's solution and the concentration of Zn ions. The pH of the solution stabilized and showed no further changes after ten days. The Zn ion release from the pure Zn specimen consistently surpassed that from bulk Zn throughout the testing duration. The concentrations of calcium (Ca) and phosphorus (P) in the solution exhibited slight reductions for both the pure Zn scaffold and bulk Zn. The unadulterated Zn scaffold experienced a weight reduction of $11.68 \pm 0.13\%$ over 60 days of immersion, yielding a degradation rate of 0.132 ± 0.004 mm/year, as shown in Fig. 17(c). The surface of the pure zinc scaffold developed a coating of white corrosion products upon immersion in Hank's solution. Over time, these substances accumulated and began to clog the scaffold's pores. SEM images of the corroded pure Zn scaffold throughout the 60-day immersion period revealed that, despite significant buildup of degradation products, its surface structure remained unaltered. EDS examination indicated that the corrosion layer on the outer struts consisted of Zn, O, P, Ca, and C. The increasing ratios of calcium to phosphorus, alongside rising levels of both elements over time, indicated a tendency for the development of calcium phosphate salts and zinc phosphate tetrahydrate. Primary degradation products detected included ZnO, zinc-containing calcium phosphate, and CaCO_3 , as shown by XRD and FTIR studies¹⁶. Fojt et al.²²¹ found that adding Zn-Ca-P as parascholzite ($\text{CaZn}_2(\text{PO}_4)_2 \cdot 2\text{H}_2\text{O}$) to biodegradable alloys effectively controlled corrosion. The thick, crystalline phosphate contact altered zinc corrosion dynamics, allowing gradual degradation while maintaining surface stability. This controlled corrosion reaction enhanced biological performance. Cells readily attached to and spread on the Zn-Ca-P-layer-coated surface in preliminary cell adhesion experiments. High-phosphate surface chemistry, favorable surface morphology, and controlled release of zinc and calcium ions improved cytocompatibility. These factors enhanced interface longevity. Zinc-based alloys met biodegradable implant requirements for initial stability, predictable deterioration, and cellular adhesion, as the Zn-Ca-P layer decoupled corrosion kinetics from early biological reactivity. This suggested that the sub-microstructured parascholzite-like layer managed material breakdown and promoted early bio-integration via a favorable phosphate interface²²¹. According to Su et al.²²², changes in pH and the accumulation of corrosion products also altered behavior in the human body. Transient Zn^{2+} or other chemical releases during rapid degradation might interact with blood components and

heighten thrombogenic risk, necessitating further investigation of the use of Zn and its alloys.

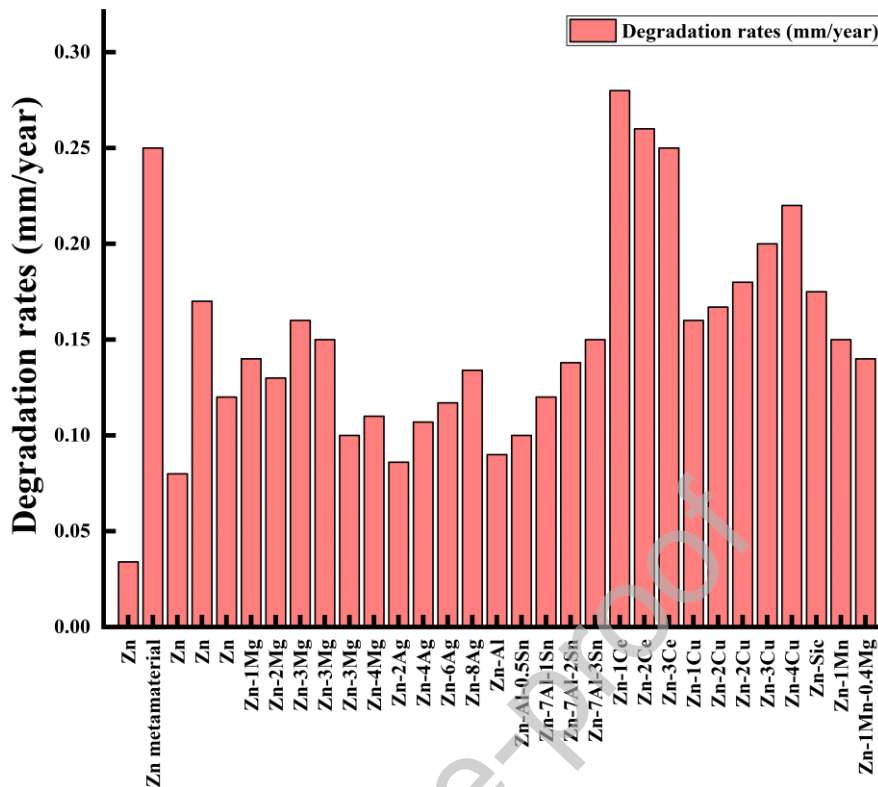


Fig. 16. Rates of degradation for pure zinc and zinc alloys (in vitro examination)

20,100,126,128,136-138,188,223-225

2.3.4 Antibacterial properties

Zinc-based alloys, including Zn-Ag, Zn-Mg, and Zn-Cu, exhibit strong antibacterial, anti-tumour, and anti-osteoporotic properties. They help prevent postoperative infection during tissue healing²²⁶⁻²²⁸. In a relevant context, Zn-Cu-based porous scaffolds exhibit significant antibacterial activity against *S. aureus* and *Escherichia coli*, with efficacy increasing with increasing Cu content²²⁹. Zn promotes anti-osteoporotic effects in regions with osteoporotic skeletal defects. Zn^{2+} regulates and catalyses various metalloproteins in the body, including alkaline phosphatase, which is closely associated with new bone development²³⁰. Zn^{2+} primarily regulates osteoblasts, osteoclasts, endothelial cells, and immune cells during bone healing to promote osteogenic differentiation of mesenchymal stem cells (MSCs), increase osteoblast bone formation, inhibit osteoclast bone resorption, stimulate angiogenesis, and induce immune regulation for new bone formation^{231,232}. Internal fixation materials based on zinc and its alloys have been developed and confirmed for their viability, exhibiting anti-osteoporotic properties²³³. According to Su et al.²³⁴, Zn ions released from Zn-based implant materials may interact with bacterial surfaces, altering cell shape and killing bacteria, thereby reducing the risk of infection and implant failure. The ZnP coating is expected to outperform other surface coatings by maintaining antibacterial activity while limiting substrate

deterioration. Samples coated with ZnP demonstrated an antibacterial efficacy of approximately 78% when incubated with *E. coli* in Lysogeny Broth (LB) medium ²³⁴.

2.3.5 *In-vitro studies*

In vitro research is conducted in a controlled environment, employs reductionist analysis, and is a cost-effective alternative to direct implantation. As shown in Fig. 17(d), cell morphology was analyzed via a laser-scanning confocal microscope across different extract concentrations. Elevated levels of Zn ions could potentially explain the reduced cell density observed in the 100% extract compared to the blank control. Cells treated with the 10% and 50% extracts exhibited significant spreading behavior and clearly defined cytoplasmic filaments. Compared to the pure bulk Zn group, the extract from the pure Zn scaffold group included approximately 1.2 times more Zn ions. Furthermore, intercellular connections resembling those observed in the blank culture control group were identified, featuring antenna-like structures. The results are consistent with the information regarding cell viability ¹⁶. Similarly, Mahdi et al. found that scaffolds were immersed in PBS solution (pH 7.4) for 8 weeks at 37°C to measure their breakdown rate. Scaffold weight loss and wall thickness decrease after PBS soaking are depicted in Fig. 17(e). Regarding weight loss, the samples lost approximately 3%-4% of their initial mass after 8 weeks of immersion in PBS. Among various designs, scaffolds with gyroid unit cells exhibited the highest average weight loss at 3.6%, while diamond scaffolds showed the lowest at 3.3%. Additionally, the results indicated that degradation behavior was influenced by unit cell size. Weight losses of 3.3% and 3.5% were observed in I-WP unit cells in sizes S01 and S02, while 3.5% and 3.3%, respectively, were recorded in Fischer-Koch S cells in sizes S01 and S03. The deterioration rate, ranging from 0.14 to 0.15 mm/year, was relatively constant for wall thickness loss across all scaffold types. The average deterioration rate for diamond, gyroid, and Fischer-Koch S scaffolds was 0.14 mm/year. The degradation rates of I-WP and primitive unit cell scaffolds, except for I-WP S01, were comparatively rapid, at 0.15 mm/year. The results were based on the degradation of scaffolds with five distinct unit-cell types over an eight-week immersion period, as shown in Fig. 17(f). By the second week, white corrosion byproducts began to accumulate around the scaffolds' pores. The unit-cell configuration influenced the formation patterns of several corrosion products. Throughout the same period, the generation of these white byproducts was more gradual in Fischer Koch S, primitive, and diamond unit cells; however, accumulation was most rapid in gyroid and I-WP scaffolds. The results indicate that the configuration of the scaffold's pores and walls influences the generation and distribution of corrosion products more significantly than the overall surface area ¹¹⁹. Importantly, in vitro findings display a critical gap compared to in vivo observations. Despite the valuable insights provided by in vitro immersion and electrochemical testing (Fig. 16), significant differences persist between degradation rates reported in the laboratory and those observed in vivo ^{235,236}. In vitro tests often demonstrate rapid and uniform corrosion of zinc-based implants; however, in vivo degradation is more gradual and inconsistent ²³⁵. The primary reason for this discrepancy is that a simple test medium cannot replicate the intricate, dynamic biological environment, in which the implant-tissue interface is continuously altered by protein

adsorption, cell activity, fluid movement, and tissue remodeling²³⁷.

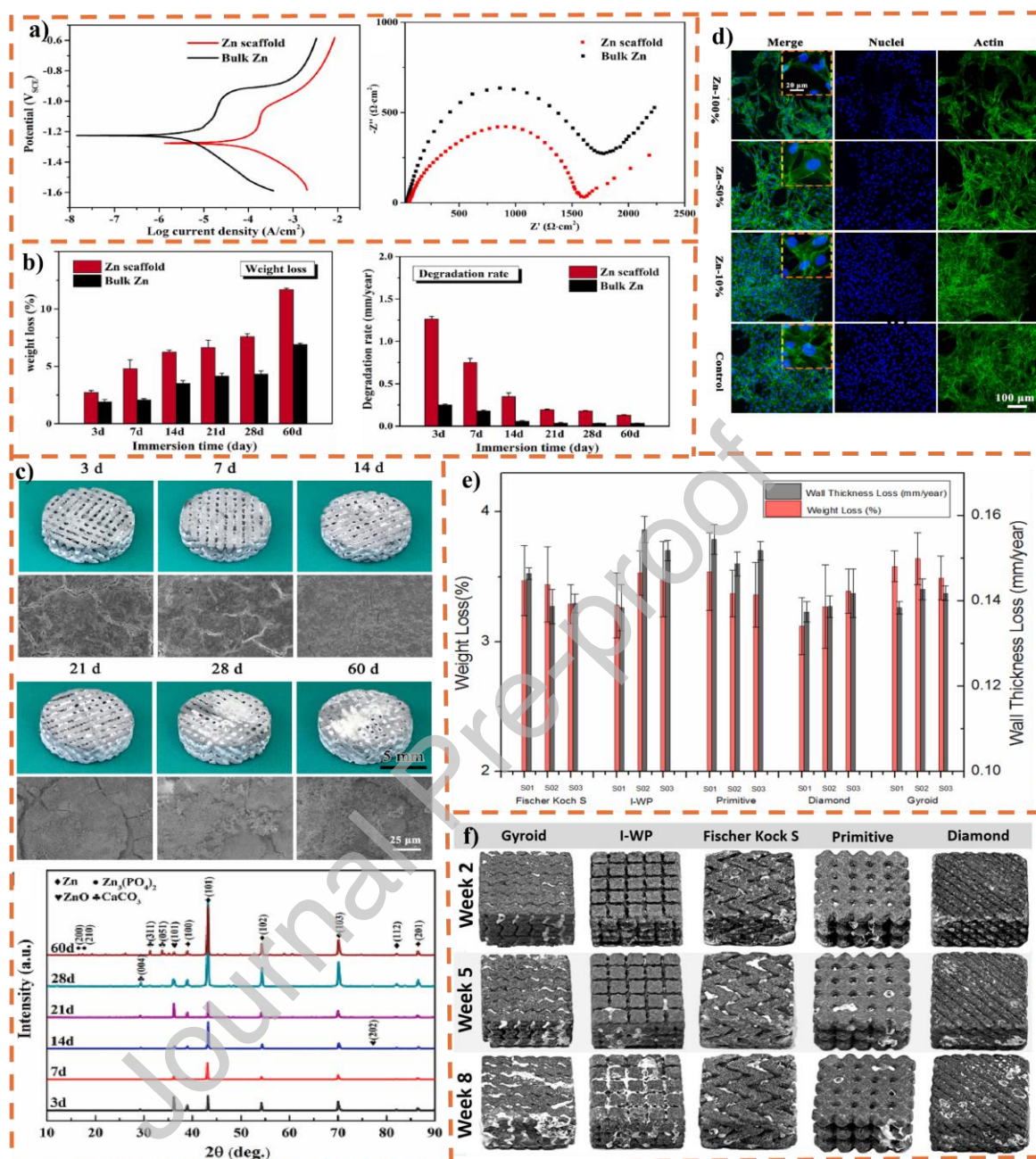


Fig. 17. (a), (b), (c), and (d) show the in vitro evaluation results of the zinc implant; (e) Demonstrates the variations in weight and wall thickness of scaffolds with three distinct unit cell sizes following an eight-week immersion in PBS and (f) Illustrates the development of white corrosion products noted after 2, 5, and 8 weeks of immersion in PBS^{16,119}.

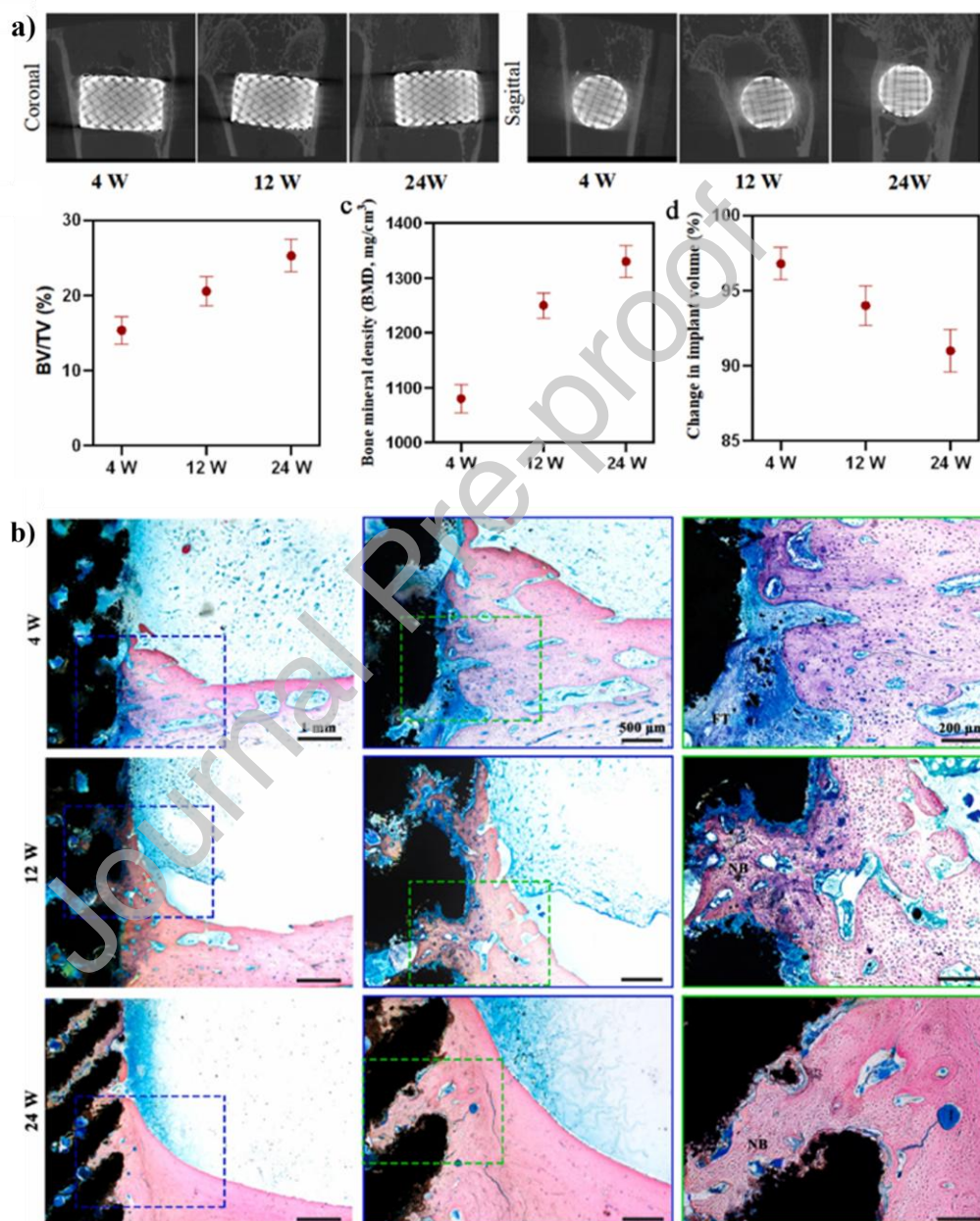
2.3.6 In-Vivo studies

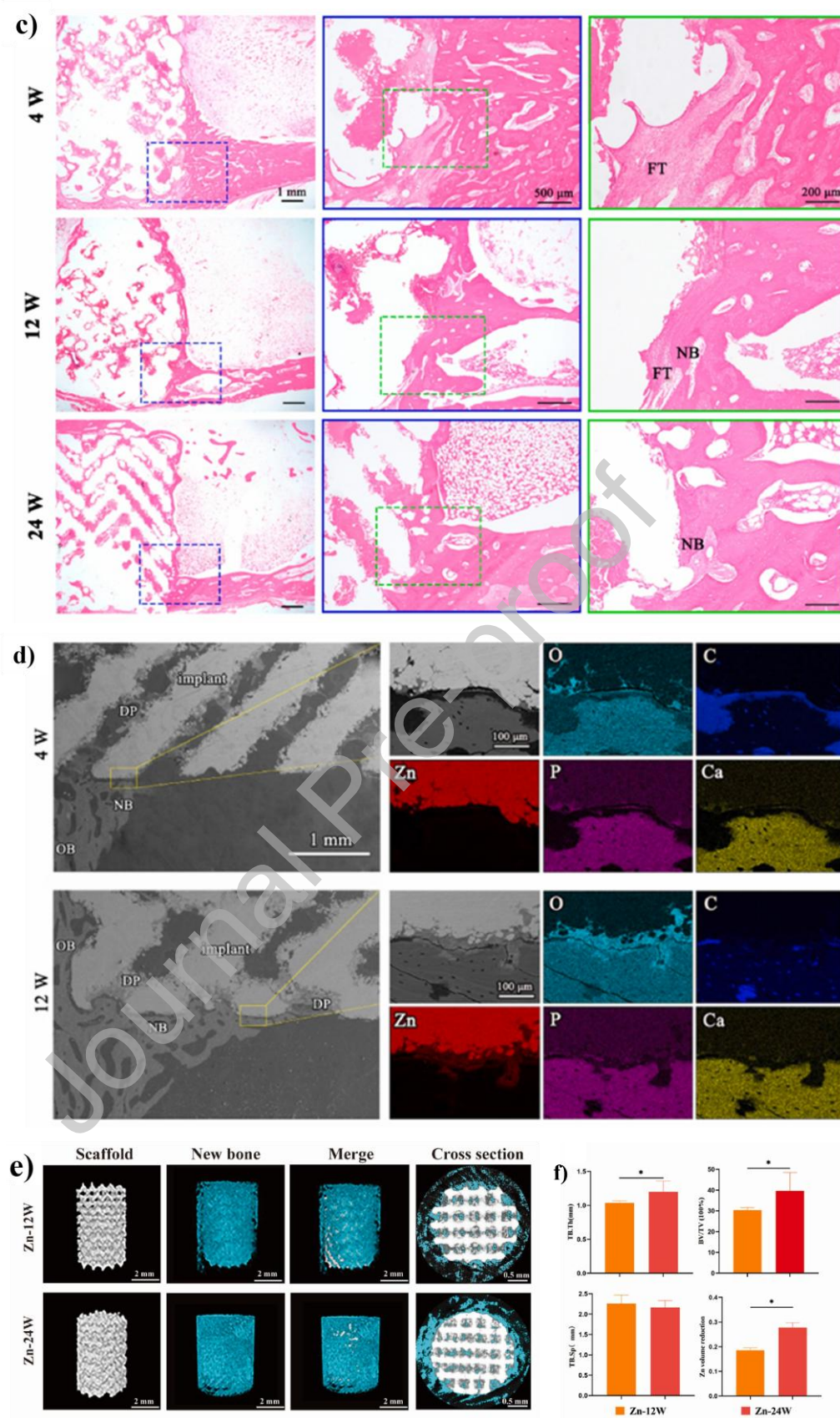
Xia D. et al.¹⁶ presented in vivo results on the rabbit femur. SEM, histological examination, and micro-CT were utilized to visualize the pure Zn porous scaffold in the rabbit femur at 4, 12, and 24 weeks, as shown in Fig. 18(a). Based on 2D reconstruction images from micro-CT

analysis, longitudinal and cross-sectional views of the scaffold were depicted. During implantation, pure zinc scaffolds exhibited no deterioration and retained structural integrity. Bone mineral density was also measured, as illustrated in Fig. 18(a). Fig. 18(b) indicates that bone density surrounding the implant increased with longer implantation duration, suggesting ongoing bone tissue formation. Concurrently, the volume of the pure Zn scaffold steadily diminished throughout the implantation period. At 24 weeks, approximately $91\% \pm 0.68\%$ of the initial scaffold volume was retained. Histological investigation using methylene blue and basic fuchsin staining revealed limited ossification on the pure Zn scaffold by the 28th day or 4 weeks, accompanied by a fibrous tissue layer separating the scaffold from the surrounding bone. By the 84th day or 12 weeks, the forming bone matrix was in direct contact with the scaffold, and bone tissue began to integrate into the porous zinc structure. At 24 weeks, increased new tissue infiltrated the scaffold pores, establishing tight contact with the pure Zn scaffold. During this period, gradual scaffold deterioration led to the accumulation of degradation byproducts and continual volume reduction. In Fig. 18(c), the existence of fibrous tissue was confirmed by the fourth week. By week 24, the previously noted fibrous tissue layer separating bone tissue from the Zn scaffold was nearly absent. No inflammatory reactions were observed throughout the monitoring period. Fig. 18(d) illustrates that SEM and EDS images of the pure Zn porous scaffold obtained at weeks 4 and 12 post-implantation showed comparable deterioration patterns. By the fourth week, a fragile layer of newly formed bone was visible on the scaffold surface; however, a gap persisted between the scaffold and adjacent bone tissue. The layer formed on the Zn scaffold comprised predominantly Zn, O, Ca, and P. By week 12, scaffolds were encased in a thicker, denser layer of newly produced bone, along with degradation products surrounding the bone tissue ¹⁶. For orthopedic applications, Yang et al. ²⁰⁰ demonstrated a biodegradable, bi-layered scaffold consisting of porous zinc and chondroitin sulfate hydrogel for the repair of an osteochondral defect. Micro-CT, SEM, and EDS were used to examine the full-thickness osteochondral defect in the Bama pig. Fig. 18(e) illustrates that micro-CT confirmed *in vivo* osteogenesis in the scaffold located in the trochlear groove. The 3D microcomputed tomography revealed that the porous pure Zn scaffolds remained in place at 12 and 24 weeks without shape alteration. Reconstructed images of the newly formed bone demonstrated osseous tissue surrounding and infiltrating the interior structure of the Zn scaffolds, with increased clarity observed in the 24-week images. Fig. 18(f) indicates that the pure Zn scaffolds exhibited a significant increase in bone volume over total volume (BV/TV) and trabecular thickness at 24 weeks compared to 12 weeks, thereby validating the 3D imaging properties. The volume of the pure Zn scaffold decreased rapidly during implantation, by approximately $18.5\% \pm 1.1\%$ and $27.8\% \pm 1.9\%$ at weeks 12 and 24, respectively. This decrease signifies an increase in bone density around the implant over time, indicating progressive bone tissue formation at varying intervals. In Fig. 18(g), new bone growth surrounding the porous pure Zn scaffolds is shown, with hard tissue slices stained with methylene blue/acrid fuchsin and Masson's trichrome. Twelve weeks post-implantation, the newly formed bone matrix was closely associated with the Zn scaffold, with fresh bone infiltrating its pores. At 24 weeks, a

greater amount of new bone was observed to have intimately attached to and developed into the pores of the pure Zn porous scaffolds. At 12 weeks post-implantation, a small amount of new bone tissue was observed in the center of the Zn scaffold. After 24 weeks, a substantial increase in new bone tissue was observed, demonstrating the *in vivo* osteogenic potential of Zn scaffolds. Fig. 18(h) presents SEM and EDS images of the porous Zn scaffolds between 12 and 24 weeks post-implantation. The pure Zn porous scaffold exhibited a consistent pattern of degradation, accompanied by a dense coating of new bone around the Zn scaffolds. The layer of degradation products covering the pure Zn scaffolds primarily consisted of Na, Zn, O, Ca, and P²⁰⁰. Recent investigations have shown that Zn alloys and Zn-doped implants degrade *in vivo*, with continuous release and accumulation of Zn ions in localized areas around bone implants. However, rapid release of Zn²⁺ may lead to excessive accumulation in local bone tissue, potentially hindering cell development and bone healing. Reports indicated that Zn²⁺ triggered apoptosis in mouse fibroblasts (NIH3T3) and mouse osteoblasts at concentrations of 70 and 400 μ M, respectively^{238,239}. Elevated Zn²⁺ ion concentrations affect BMSC apoptosis, though the underlying mechanisms remain unclear. These issues are crucial for understanding how zinc microenvironments influence BMSC development and osteogenesis. Apoptosis is a gene-regulated, self-directed form of cell death distinct from cell necrosis. Cytotoxicity and intracellular ROS induce apoptosis in rBMSCs. Recent research suggests that excess metal ions, such as iron, can induce oxidative stress by increasing ROS and promoting apoptosis and pyroptosis²⁴⁰. This suggests that ROS may play a significant role in metal-ion-induced apoptosis. Zn²⁺ microenvironments were examined for their effects on rBMSC ROS production. Fluorescence labeling and flow cytometry analysis indicated that intracellular ROS generation in rBMSCs increased with Zn²⁺ concentration. High ROS levels induced by metal ions such as Cu and Ag inhibited hBMSC osteogenic activity²⁴¹. Consequently, elevated ROS levels in rBMSCs explain the negative impact of high Zn²⁺ ion concentrations on the osteogenic differentiation of rBMSCs²⁰⁷. These studies illustrate the need to carefully regulate Zn²⁺ addition and removal from zinc-incorporated biomaterials. The optimal Zn²⁺ microenvironments for BMSC osteogenic development were established. Zn²⁺ ions influence BMSC osteogenic activity in two ways. The dual regulation of Zn²⁺ microenvironments on BMSCs must be fully understood to maximize their biological benefits. Extracellular Zn²⁺ does not immediately affect BMSC osteogenesis; transmembrane zinc transporters on the cell surface must internalize and store it at a specific level before physiological effects occur²⁴². Transmembrane zinc transporters regulate cell Zn²⁺ levels. The ZIP (Zrt- or Irt-like protein, SLC39) family serves as the primary mechanism for zinc entry into cells. By contrast, zinc exporters from the ZnT (cation diffusion facilitator, SLC30) family reduce cytoplasmic zinc levels. Thus, these two mechanisms regulate cell Zn²⁺ levels. ZIP1 protein expression and zinc absorption increase during pluripotent MSC osteogenic differentiation into osteoblasts. Bi-directional zinc transport is necessary for BMSC osteogenic differentiation²⁴³. Zn²⁺ microenvironments influence BMSCs by promoting adhesion and proliferation (2-5 μ g mL⁻¹), modestly elevating intracellular Zn²⁺ content via zinc transport, and facilitating osteogenic

differentiation through the MAPK/ERK signaling cascade. However, excessively high levels may be detrimental and warrant further investigation²⁰⁷. According to Li et al.²⁴⁴, the L-PBF of Zn and Ti6Al4V supports the growth of MG63 cells on scaffolds with different lattice or unit cell types, surface treatments, and immersion medium volumes. They demonstrated that unit-cell selection is crucial for biocompatibility, cell attachment, and cell growth. The unit cell exposed surface area varies in contact with blood or human cells, resulting in differing Zn^{2+} absorption concentrations.





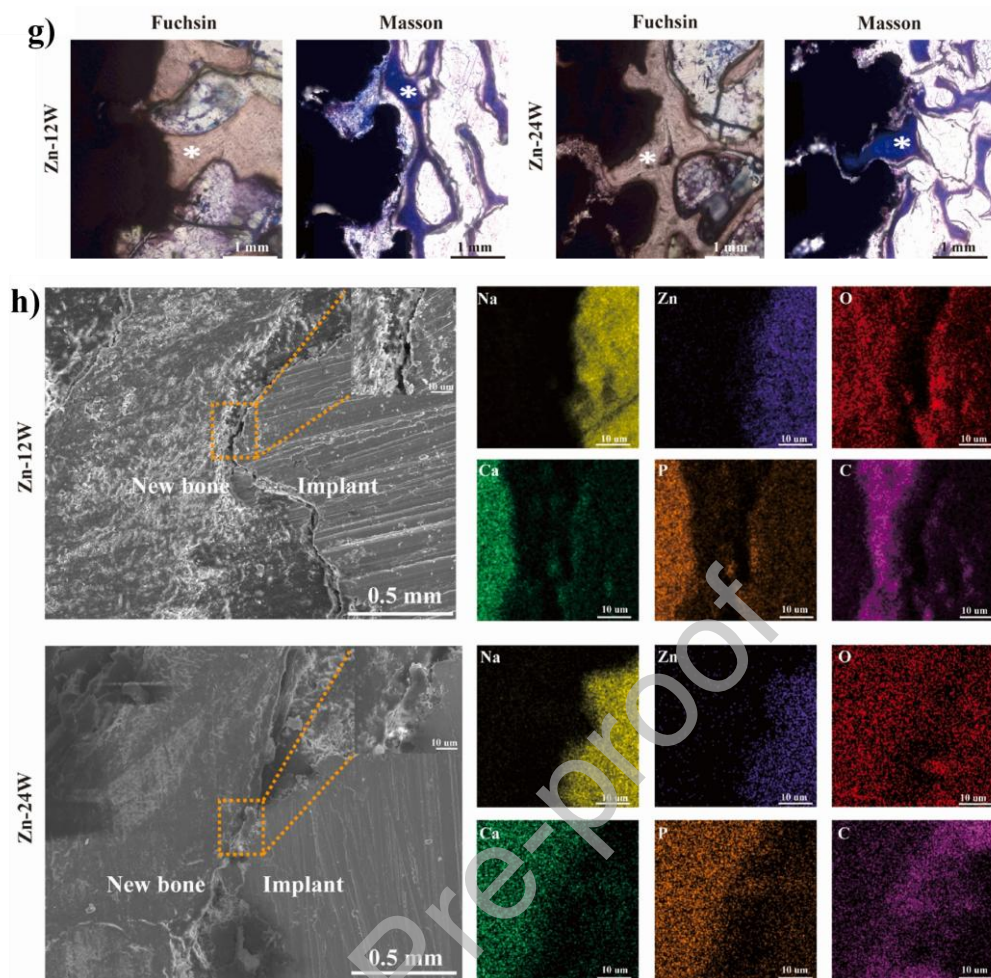


Fig. 18. (a), (b), (c) and (d) In vivo results of the rabbit femur defect on the zinc implant ¹⁶; (e), (f), (g) and (h) In vivo results of the Bama pig of osteochondral defect on the zinc implant after 12 and 24 weeks ²⁰⁰.

3. Lattice structures for biomedical implants

The attributes of unit cells, including unit type, porosity, and size, are essential for assessing the dependability and efficacy of porous implants made from various unit cells. Thus, the choice of unit cell in scaffold design requires careful assessment, as it directly influences stress distribution and the scaffold's mechanical properties ²⁴⁵. The primary methods for constructing unit cells are parametric and non-parametric designs ²²⁴. CAD techniques are primarily employed for non-parametric designs, also known as geometry-based designs. The diamond and its corresponding unit cells exemplify non-parametric cells commonly utilized in bone implants. A diamond unit exhibiting 67% porosity was present in the initial L-PBF pure Zn scaffolds [103]. Li et al. produced pure Zn specimens or scaffolds featuring diamond units with varying porosity distributions ²⁴⁶. The lattice structures of zinc and its alloys exhibit distinct mechanical properties, as shown in Fig. 19. Additionally, in conjunction with diamond units, non-parametric unit cells such as the dodecahedron, octet truss, FCC, and 3D Kagome were utilized to construct pure Zn scaffolds ⁶³. The non-parametric design, while simple and well-analyzed, is poorly suited for orthopaedic applications due to abrupt transitions between struts,

which result in stress concentrations ²⁴⁷. Parametric design, which produces cells using algorithms, can enhance structural design. The TPMS method is the most frequently mentioned parametric design technique used in bone implants. Benefits of TPMS-based unit cells include high permeability, smooth transitions, and exceptional pore connectivity. Afshar et al. ²⁴⁸ report that smooth transitions contribute to more consistent stress distribution in scaffolds during compression. Reduced stress concentration accounts for improved mechanical characteristics of TPMS unit cells compared to other unit cell types ²⁴⁸. Furthermore, the increased surface area of the scaffold may impact degradation and the cell function-to-volume ratio, necessitating the selection of appropriate pore size and shape. In addition to optimized mechanical behavior, TPMS units can mimic native bone tissue, integrating with surrounding tissues and supporting physiological functions, including nutrient transport, oxygen diffusion, and ion exchange ²⁴⁹⁻²⁵¹. Zn-based BM scaffolds for orthopaedic implants have been designed and manufactured using TPMS devices such as the gyroid and IWP. The TPMS structure's large relative surface area, enhanced permeability, and improved cell adhesion and retention are noteworthy. According to Karageorgiou et al., cell development and migration require hole diameters of at least 100 μm . Holes larger than 300 μm can enhance blood vessel formation and cell proliferation, as shown in Fig. 20(a) ²⁵². Additionally, gyroid unit-directed curvatures at the cellular level increase cytoskeletal contraction and extend cell shape, thereby improving osteogenic outcomes ^{17,253}. Scaffolds constructed from Zn-0.8 Li-0.1 Mg have been reported by Liu et al. ¹³⁴ with porosities of 60, 70, and 80%. "Architected metamaterials," structural designs with unique characteristics that can be precisely designed and adjusted, may offer favorable combinations of features required for implantation. However, their advanced structural complexity complicates the manufacturing of architected metamaterials in three dimensions. Structures based on Voronoi principles consist of connected polyhedra formed by perpendicular bisection planes connecting neighboring seeds, corresponding to the primary histomorphometric indices of bone trabeculae. Their ideal characteristics for cell penetration, nutrient transfer, osteo-conduction, and cell proliferation render them well-suited as engineered metamaterial options for trabecular bone implants. Significant topological control potential is demonstrated by the mechanical performance of additively manufactured Voronoi-based structures, enabling control of elastic modulus and achieving exceptional energy absorption for human implants, as illustrated in Fig. 20(b) ^{103,254,255}. Recently, Shi et al. ¹⁸⁸ developed negative Poisson's ratio (NPR) metamaterials that exhibit advantages due to their bone-mimicking properties and energy absorption, as shown in Fig. 20(c). This 3D-optimized reinforced unit cell retains NPR properties while significantly improving scaffold strength. L-PBF successfully developed and produced 3D auxetic Zn meta-biomaterials. Internal and exterior reinforcing components were added to enhance mechanical characteristics, and the Poisson's ratio and overall mechanical performance of these structures were adjusted by varying strut diameters and concave angles. Compared to traditional cubic structures, it meets the requirements of cancellous bone and exhibits effective energy absorption. This also indicates that increased permeability in this metamaterial accelerates degradation, with the scaffold losing 19.76% of

its weight over 28 days while increasing its yield strength by 2.3 MPa. Maintaining NPR characteristics, regulated nutrition, metabolite and cellular response trafficking, and a suitable degradation rate are crucial²⁵⁶. Bone replacements experience multi-directional stress in vivo, with natural bone tissues typically being anisotropic and hierarchical^{257,258}. Orthopaedic implants confront challenges in complex and dynamic physiological conditions. These issues include material degradation, changes in stress distribution, and growth stimulation during the bone-healing process. In situations with multi-directional stress, isotropic structures generally respond poorly to fluctuating stress levels, leading to tension buildup in certain areas²⁵⁹. As material breakdown occurs and trabecular bone forms, the mechanical properties of bone scaffolds, such as yield strength, must adapt to changing physiological requirements²⁶⁰. According to FEM analysis, NPR constructions demonstrated a uniform stress distribution and a reduction in stress concentration of up to 48.89%. During elastic compression, Poisson's ratios ranged from -0.56 to 0. In compression testing, the structure with the highest elastic modulus and yield strength of 1327.17 and 15.3 MPa, respectively, exhibited a porosity of 61.98%. These values met the mechanical performance criteria of human cancellous bone and were comparable to those of a basic cubic structure with corresponding porosity¹⁸⁸. L-PBF Zn NPR scaffolds exhibit significant potential as bone scaffolds due to their exceptional mechanical properties and favorable degradation behavior.

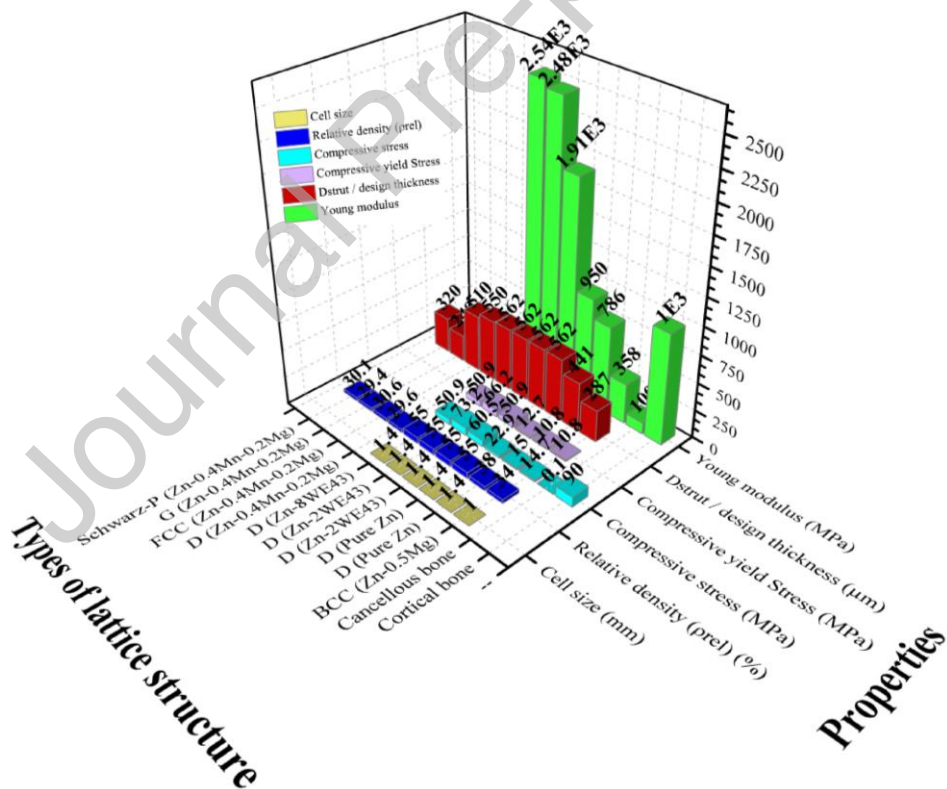


Fig. 19. Lattice structures for the biomedical applications of zinc and its alloy, body-centred cubic unit cell (BCC), diamond (D), primitive (P), and gyroid (G)^{73,110,176,261,262}.

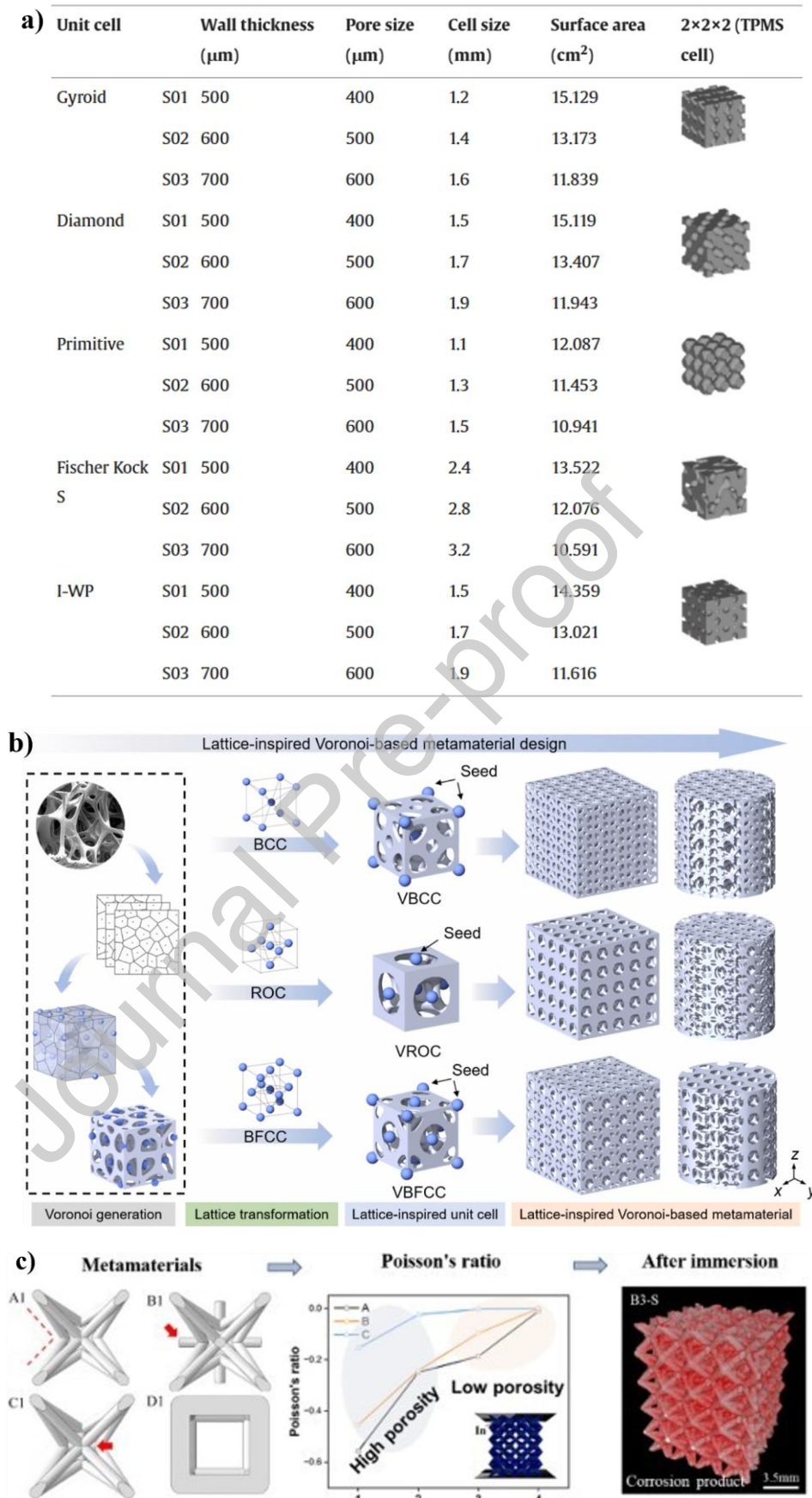


Fig. 20. (a) TPMS zinc structures for the biological implants; (b) Designed architecture of Voronoi-based metamaterial models for zinc that are inspired by lattices and use different

lattice configurations, such as Voronoi generation, lattice transformation, and lattice-inspired unit cell production, constructing metamaterials that are based on Voronoi and lattice^{103,254,255}; (c) NPR metamaterial for zinc-based biomaterial¹⁸⁸.

4. Applications of zinc and its alloys

The most commonly used biodegradable metals are employed in orthopaedic fixation devices, degradable cardiovascular stents, wound sutures, GBR, and biodegradable energy storage (battery) systems, which rely on biocompatible and biodegradable materials instead of toxic or non-degradable substitutes. Natural biodegradable materials and substrates, along with the encapsulation of items in biocompatible macromolecular polymers, have been utilized. These materials include silk protein, spider silk, cellulose, polylactic-co-glycolic acid (PLGA), polycaprolactone, and poly(1,8-octane diol-co-citrate)^{263,264}. Comprehensive studies indicate that thin films or foils may dissolve water-soluble metals, including zinc, magnesium, and manganese. These metals are applied in biodegradable electronic devices as bioresorbable current collectors or circuits²⁶⁵, biomedical plate and screws (orthopaedic fixation devices)²⁶⁶, biodegradable cardiovascular stents [268], and wound sutures and GBR²⁶⁶, as shown in Fig. 21.

4.1 Biomedical plate and screws (orthopaedic fixation devices)

Biodegradable metals are suitable for medical applications due to their rapid degradation, strong mechanical properties, and safety for living organisms. A recent study has identified zinc as a promising candidate, gaining significant attention. Unlike magnesium, zinc corrosion does not produce hydrogen gas pockets, which can impede bone healing, induce cortical abnormalities, and promote callus formation²¹⁵. Since Zn is the most important mineral for humans, Zn alloys exhibit excellent potential as biodegradable components for implantable devices. Pure Zn is brittle and soft, lacking mechanical strength. By contrast, Zn alloys demonstrate improved mechanical properties while maintaining the advantages of a gradual release profile and suitable biological compatibility. Regarding biocompatibility, significant advantages have been noted. Indirect cytotoxicity and cell adhesion experiments indicate that the Zn alloy extract medium supports appropriate cell morphologies, tolerance, and attachment. These alloys can be utilized in orthopedic implants, orthopedic fixation devices, and cardiovascular stents^{94,267}. Recent findings show that pure Zn screws were more effective in reducing bone tunnel enlargement (BTE) following anterior cruciate ligament (ACL) repair in rabbits than pure Mg and Ti screws, as illustrated in Fig. 21 (a)²⁶⁶.

4.2 Biodegradable batteries

Biodegradable main battery systems provide a practical power source for bioresorbable electronic medicine due to their high energy density and straightforward insertion, eliminating the need for additional surgeries for retrieval. Currently, biobatteries used as temporary implants face challenges regarding operational lifespan, compatibility with biological systems, and natural breakdown, which limit therapeutic potential. Zn is a compelling anode material because of its low equilibrium potential of -0.763 V relative to the standard hydrogen electrode,

high hydrogen evolution potential, reduced unfavorable reactions, and an estimated volume-specific capacity of up to 5822 mAh cm⁻³. Zinc is essential for various biological functions, including bone formation, cognitive function, and regulation of endocrine balance ^{268,269}. Compared to existing primary Zn biobatteries, a biodegradable primary zinc–molybdenum (Zn–Mo) battery exhibits enhanced energy capacity and output voltage, capable of operation for up to 19 days. The Zn–Mo battery system's exceptional compatibility with biological systems and natural degradation capacity significantly enhance the growth and multiplication of dorsal root ganglion cells and their axons. The biodegradable battery module, consisting of four Zn–Mo cells connected in series, employs a gelatin electrolyte to facilitate the electrochemical production of signaling molecules (nitric oxide, NO) and achieves efficacy comparable to traditional power sources, as shown in Fig. 21(b) ²⁷⁰.

4.3 Bio-degrading cardiovascular stents

Numerous zinc-based products for use in the human body have arisen due to zinc's significant function. Several zinc alloys show promise as vascular stent materials ²⁷¹. Zinc is essential for maintaining healthy heart function, and deficiency can lead to cardiovascular disease ²⁷². Serum Zn levels above 100 µg.dL⁻¹ are linked to an increased risk of cardiovascular disease and coronary heart disease; thus, maintaining adequate levels in the body is important, even though Zn has several health benefits ²⁷³. In a study where a pure zinc stent was placed in a rabbit's abdominal aorta, persistent corrosion of the stent was observed prior to endothelialisation. Localised corrosion is the main reason for stent degradation after endothelialisation ²⁷⁴. To maintain the mechanical integrity of vascular stents, degradation rates of Zn stents must be modified so that they degrade uniformly and localised corrosion is minimized, as shown in Fig. 21 (c) ²⁷⁵.

4.4 Wound sutures and guided bone regeneration (GBR)

Zn-based BMs are superior to polymers in addressing these issues due to their strength and hardness, allowing staples to be made thinner and occupy less space. Polymer staples have demonstrated efficacy in wound closure; however, their strength is inadequate, resulting in wound reopening at surgical incisions. Additionally, the rigidity of polymers is insufficient for thicker tissues. The limited space beneath the skin complicates incision procedures in the arms and legs, as slight inversion of the skin is essential post-staple application. New Zn alloys with superior mechanical properties may be combined with other elements to manufacture surgical staples. Biocompatible alloying elements Mg, Ca, Mn, and Cu were selected for their ability to maintain physiological homeostasis and promote early healing ²⁷⁶. The degrading behavior of Zn in an artificial saliva solution was analyzed, along with its cytotoxicity and antibacterial activity, to assess its viability as a GBR membrane. Zn has been shown to effectively kill *Porphyromonas gingivalis*. Its alloys may significantly reduce the number of germs adhering to surfaces, forming colonies, and creating biofilms, as illustrated in Fig. 21(d) ²⁷⁷.

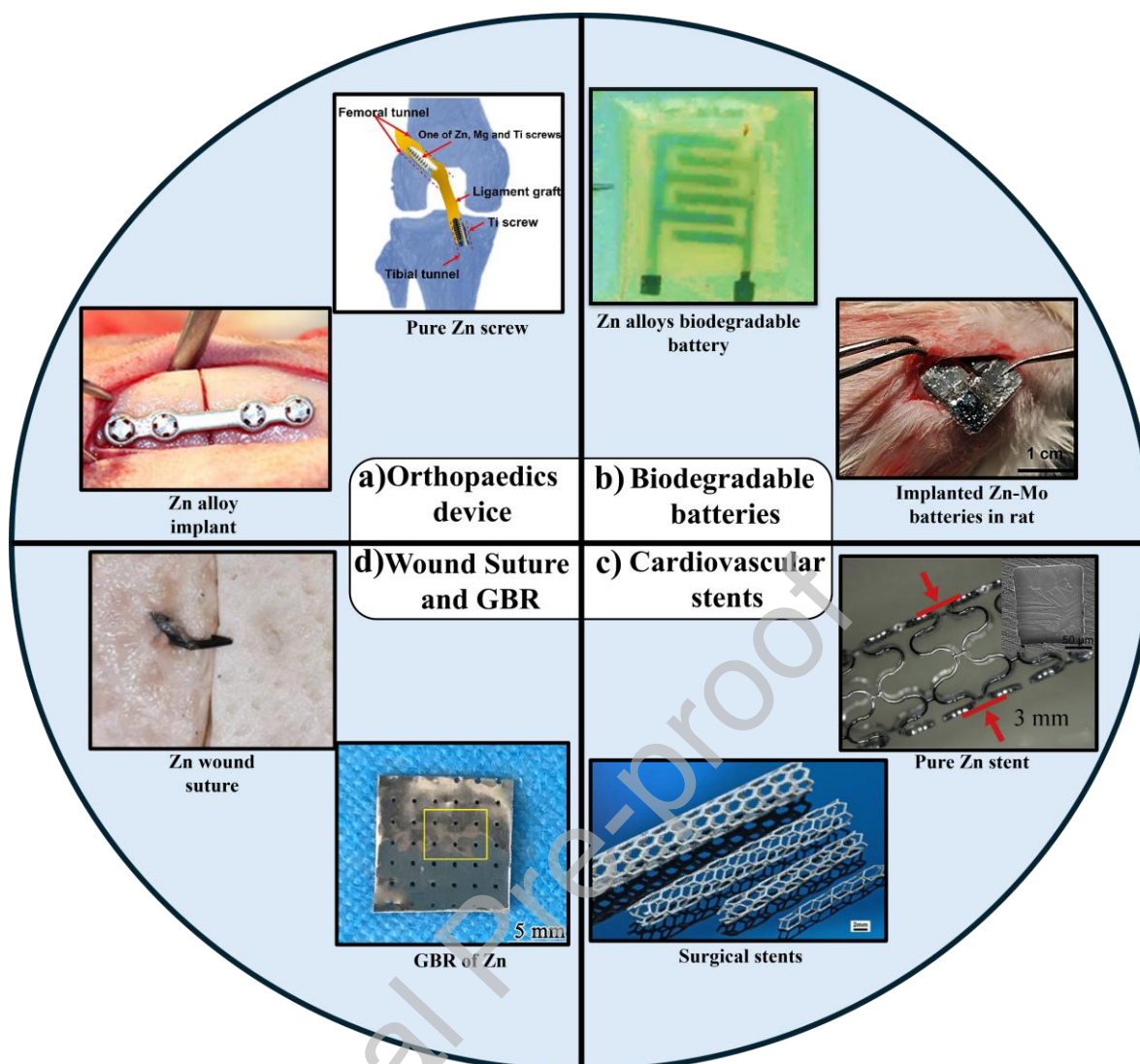


Fig. 21. (a) Application of Zinc and its alloys; (a) Orthopaedic device^{56,276,278}, (b) biodegradable batteries²⁷⁰, (c) Degradable Cardiovascular stents²⁷⁴, (d) wound sutures and GBR^{279,280}.

5. Challenges and their future perspectives

The evaporation of pure zinc presents challenges for deposition processes, necessitating an environment free from oxidative conditions and thoroughly purified. Given the distinct requirements for medical applications, the production of zinc-based BMs exhibiting enhanced mechanical strength, suitable degradation properties, improved biocompatibility, and diverse bioactive functions, including pronounced angiogenesis and anti-cancer properties, is essential. The artificial intelligence (AI) and machine learning (ML) process serves as a rapid approach for predicting material quality and identifying combinations with the most significant attributes. The structural design of zinc-based BM scaffolds significantly influences their overall effectiveness. The consequences and essential processes of structural design on the biocompatibility and in vivo degradation behavior of Zn-based biomaterials remain inadequately explored. Further investigation of these findings is vital, as they contribute

significantly to the development of scaffolds with specialized performance. Additionally, topological optimization and machine learning are crucial in developing the ideal scaffold structure, ensuring precise control over the mechanical and biodegradable properties of implants. In the early stages of development, modifications to the surface of L-PBF Zn-based BM scaffolds will be the primary focus. The application of coatings, particularly hydrogels, on scaffolding surfaces presents novel options. The use of these coatings substantially reduces the discharge of excess Zn^{2+} , thereby enhancing biocompatibility. Moreover, these coatings may function as drug transporters, rendering the scaffolds bioactive. Currently, orthopaedic implants represent the primary focus of research concerning L-PBF Zn-based biomaterials. The utilization of L-PBF is becoming significant in the fabrication of therapeutic devices due to its capacity to produce intricate structural designs. Examples include biodegradable energy devices, interbody fusion devices, cardiovascular stents, and guided bone regeneration membranes. Zinc-based biomaterials may be engineered and fabricated by L-PBF to address a broader range of therapeutic applications, considering the distinct functional requirements of these applications. Thorough testing is essential in both *in vitro* and *in vivo* settings to ensure effectiveness and safety for therapeutic applications. Important future scopes are discussed below.

5.1 Long-term or extended biodegradation

Zn-based biomaterials undergo physiological corrosion, and the *in vivo* corrosion mechanisms of Zn-based biomaterials have been thoroughly investigated. However, certain essential elements remain ambiguous. Human interstitial fluid, the principal bodily fluid post-implantation, contains proteins, carbohydrates, amino acids, and inorganic ions such as chloride and carbonate. This fluid also contains buffers. Less than 5% of the bone microenvironment exhibits physiologically low oxygen levels, suggesting a correlation between oxygen consumption and zinc breakdown. Understanding the influence of these physiological variables on the deterioration of Zn-based implants represents a significant advancement in the field. The susceptibility of Zn-based implants to internal healing in fractures induced by physiological fluids, such as stress corrosion cracking, may jeopardize their mechanical integrity. During prolonged deterioration, the corrosion fatigue of the porous Zn-based scaffold severely affects the mechanical integrity of the bone tissue engineering (BTE) scaffold. Crevice-induced corrosion may also weaken Zn-based implants²⁸¹.

5.2 Response of the host or patient

In vivo animal studies have elucidated the molecular mechanisms underlying zinc-based biomaterials, demonstrating the absence of systemic adverse effects. The long-term biosafety of degradation products must be assessed, encompassing absorption, distribution, metabolism, and excretion. The leakage of Zn ions from orthopaedic implants must be evaluated during extended reduction periods. This comprehensive study is essential for assessing the long-term implications of Zn-based implant failure on human health and safety. Initial degradation products, such as zinc and hydroxide ions, have been investigated in biological systems. Zn-based bone implants are designed to degrade into various chemical species in a physiological

environment²⁸². The primary biologically active intermediate compounds found on the surface or subsurface of zinc-based implants include zincite, zinc hydroxide, and zinc phosphate. Degradation products, regardless of solubility, may activate the host's adaptive or innate immune systems²⁸³. Given recent breakthroughs in evidence-based biological materials and medical studies, standardized requirements for the biological security of Zn-based materials should be broadened and validated²⁸⁴.

5.3 Material optimisation

Zinc ions are primarily responsible for the bioactivity and functionality of zinc-based implants, including osteogenesis, vascular development, immune system regulation, and antibacterial properties. This may stimulate the proliferation of fibrous encapsulation around the implant, facilitating new bone formation. Distinctive design modifications of zinc alloy implants and composite/alloying are necessary for the regulated disintegration or corrosion of zinc metal and the subsequent release of zinc ions. Various chemical modifications, including micro-arc oxidation, anodic oxidation, and chemical deposition, alongside physical modifications such as magnetic sputtering, dip coating, and mechanical surface treatment, have significantly enhanced biodegradability and biocompatibility²⁸⁵.

5.4 Machine learning-guided process optimisation

Optimisation of newly developed L-PBF Zn alloys often involves extensive trial-and-error. AI and ML may be utilized to improve L-PBF process parameters for the Zn alloy. A supervised Gaussian Process Regression (GPR) approach was employed to identify the optimal high-density process window. The combination of high-power and low-speed (HPLS) characteristics, which is infrequently used, requires investigation. Forecasting of microstructures, mechanical characteristics, degradation performance, biological properties, and antibacterial properties is also feasible. By advancing machine learning algorithms, predictive models can enhance the efficiency of optimising the AM process. This would facilitate the development of more effective and targeted biodegradable metals¹³⁶.

5.5 AI/ML-induced computational Zn alloy composition

One main strategy to improve the performance of biomedical implants involves refining the alloy composition by controlling the microstructure²⁸⁶. The current empirical-guided approach for producing Zn alloys is inefficient and time-consuming, resulting in high costs. Machine learning, an emerging method in scientific research, has quickly become a tool for locating and creating new materials^{287,288}. Research on biodegradable Zn alloys using machine learning faces challenges due to insufficient high-quality data¹³⁶. Recently, two machine learning models were developed by Chen et al.²⁸⁹ to construct biodegradable Zn-Mn alloys significantly faster than the traditional trial-and-error approach. These models focus on a biodegradable Zn alloy. The computational alloy composition of biodegradable Zn alloys is designed to achieve a 400 MPa grade through machine learning. The prediction errors for its UTS and CR are only 0.95% and 5.5%, respectively, as verified by experiments. This outcome may facilitate future applications of AI/ML in alloy composition by reducing errors and increasing the available training data, thereby enhancing the computational method's efficiency.

Data availability

The data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

CRedit authorship contribution statement

Aditya Nigam: Methodology, Writing – original draft, data curation, investigation. **Satvasheel Powar:** Supervision, Writing – review and editing. **ASS Balan:** Visualization, Supervision, Writing – review and editing, Conceptualization. **Mrityunjay R. Doddamani:** Supervision, Writing – review and editing, conceptualization, Visualization.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: